EPI DIGITAL SCHOOL

LittleBig Connection

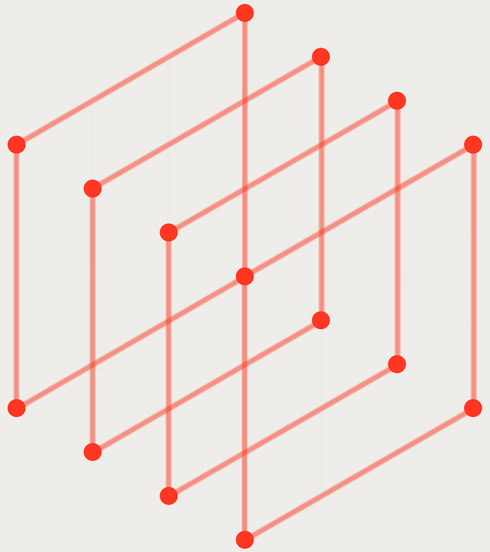
databricks

Ministry of Higher Education and Scientific Research

End-of-Studies Internship

Architecting and Operating a Unified Data Lakehouse on Azure Databricks

Building a Scalable and Governed Foundation for Enterprise AI and Data Science



Presented by: Ahmed Hmila

Supervised by: Dr. Takwa Ben Aicha

Ms. Jebrane Narjess

Mr. Raouf Haddada

Data is everywhere





On average, how much data does a single person generate per year?

A. Less than **100 GB**

B. Between **100** and **200 GB**

C. Between **200** and **300 GB**

D. Between **500 GB** and **1 TB** *

* Digital Family Insights — How much data can one person generate?
digitalfamilyinsights.com
DesignRush — How Much Data is Created Every Day
designrush.com

Data is the new Gold

Role of the Data Engineer



Data

Asset

Planning

1. Introduction to Little Big Connection and its services

2. Architecture and key Components

3. Data Operations

4. Value and Impact

5. Conclusion & Perspectives



Planning

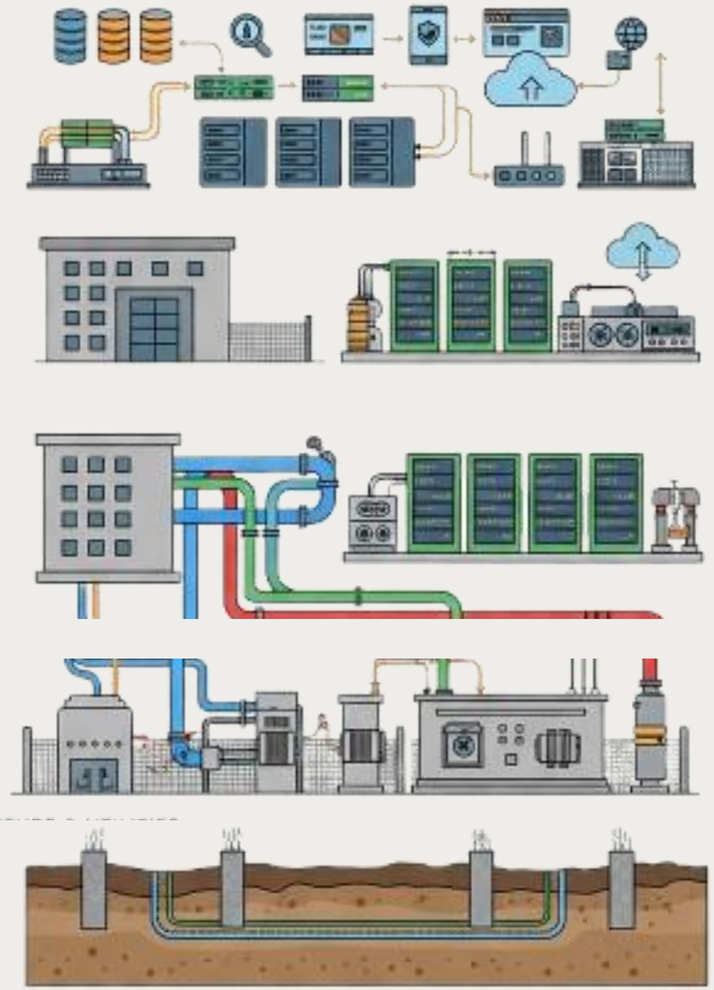
1. Introduction to Little Big Connection and its services

2. Architecture and key Components

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4. Value and Impact

5. Conclusion & Perspectives



Planning

1. Introduction to Little Big Connection and its services

2. Architecture and key Components

3. Data Operations

4. Value and Impact

5. Conclusion & Perspectives



LittleBig Connection

Founded
2013

Joined

201
5

mantu

mantu

Independent international consulting firm supporting business transformation.

Mantu' Clients



Mantu' Profits

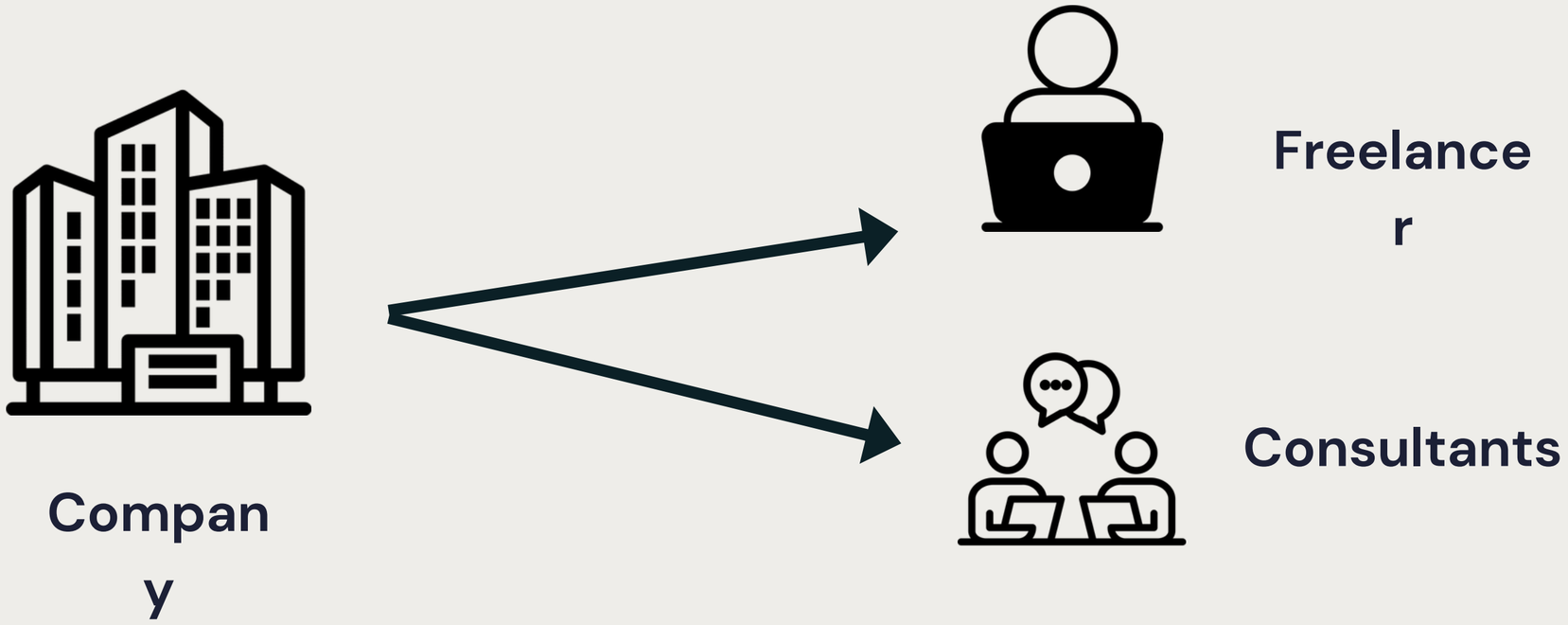
(2024)

1 Billion €



1. Introduction to Little Big Connection and its services

LittleBig Connection



Create a clear, direct link between **large companies** and a global network of experts, **freelancers**, and **consultants**.

LB's Clients:


LVMH
Opella.
Valeo
Lyreco
ENGIE
LittleBig Connection
NESPRESSO
Air Liquide
creative oxygen
bouygues TELECOM
AXA

Eleven VMS

Company

Manage

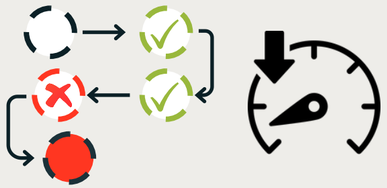
Platform to :

Projects

Suppliers

- Handle RFPs (Request for Proposal) “Job offer”
- Track projects
- Communicate efficiently

Initial State of Eleven VMS



Inefficient Workflow



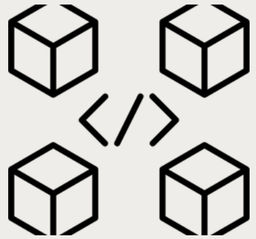
Lack of Scalability



Poor Data Governance



Inconsistent Data



Fragmented Data

Initial State of Eleven VMS



Inefficient Workflow



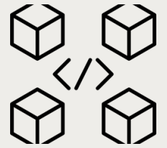
Lack of Scalability



Poor Data Governance



Inconsistent Data



Fragmented Data



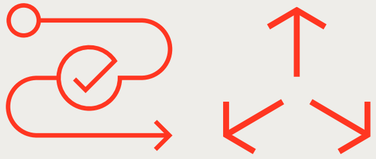


Unreliable Data Insights



NO AI Capabilities

Objectives for Eleven VMS



Efficient
Scalable
Pipelines



Unified
cloud-native
Data Lakehouse



Automated
ETL
Process



Secure
governed
database

2. Architecture and key Components

Whats GDPR ?

(General Data Protection Regulation)

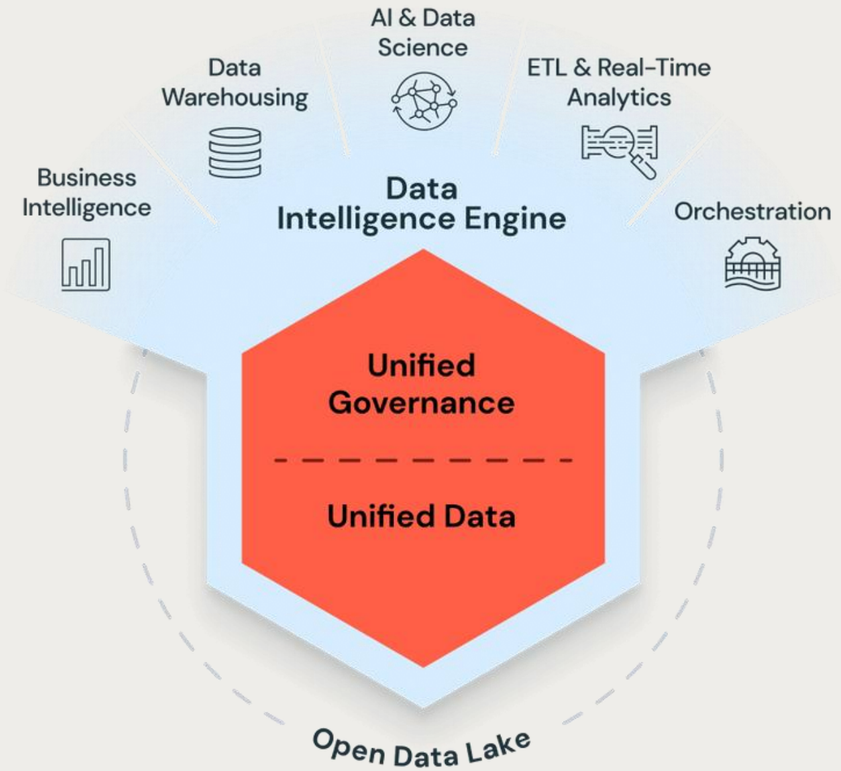
- Protect the personal information of EU citizens
- Mandatory for companies operating within the EU

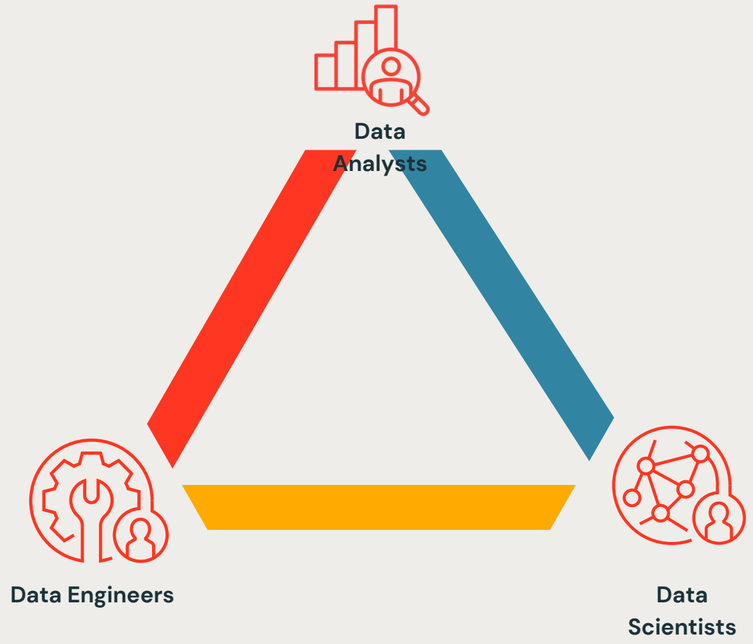
- Core design principle in our project



databricks

All-in-one high-tech workshop for data





Tools For Everyone

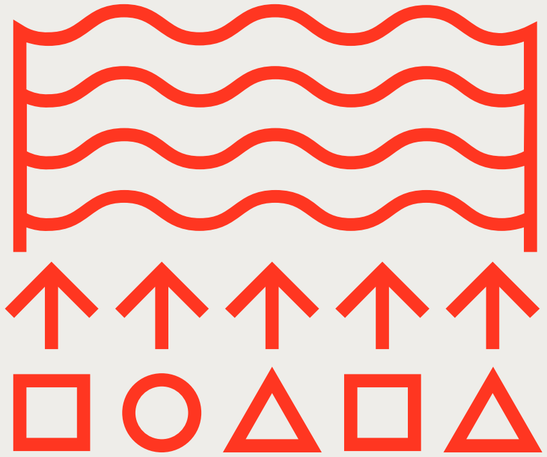
Where should the data go in the cloud ?

Data Warehouse



Vs.

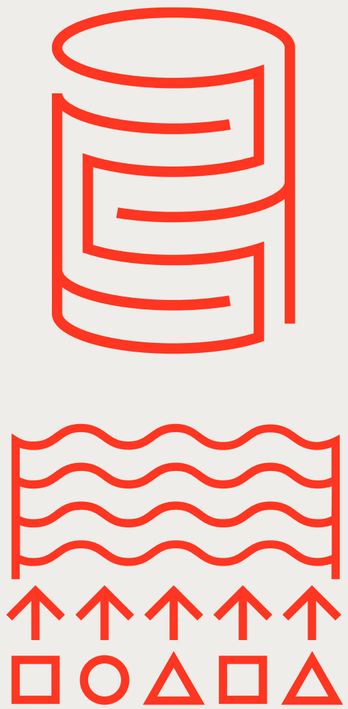
Data Lake

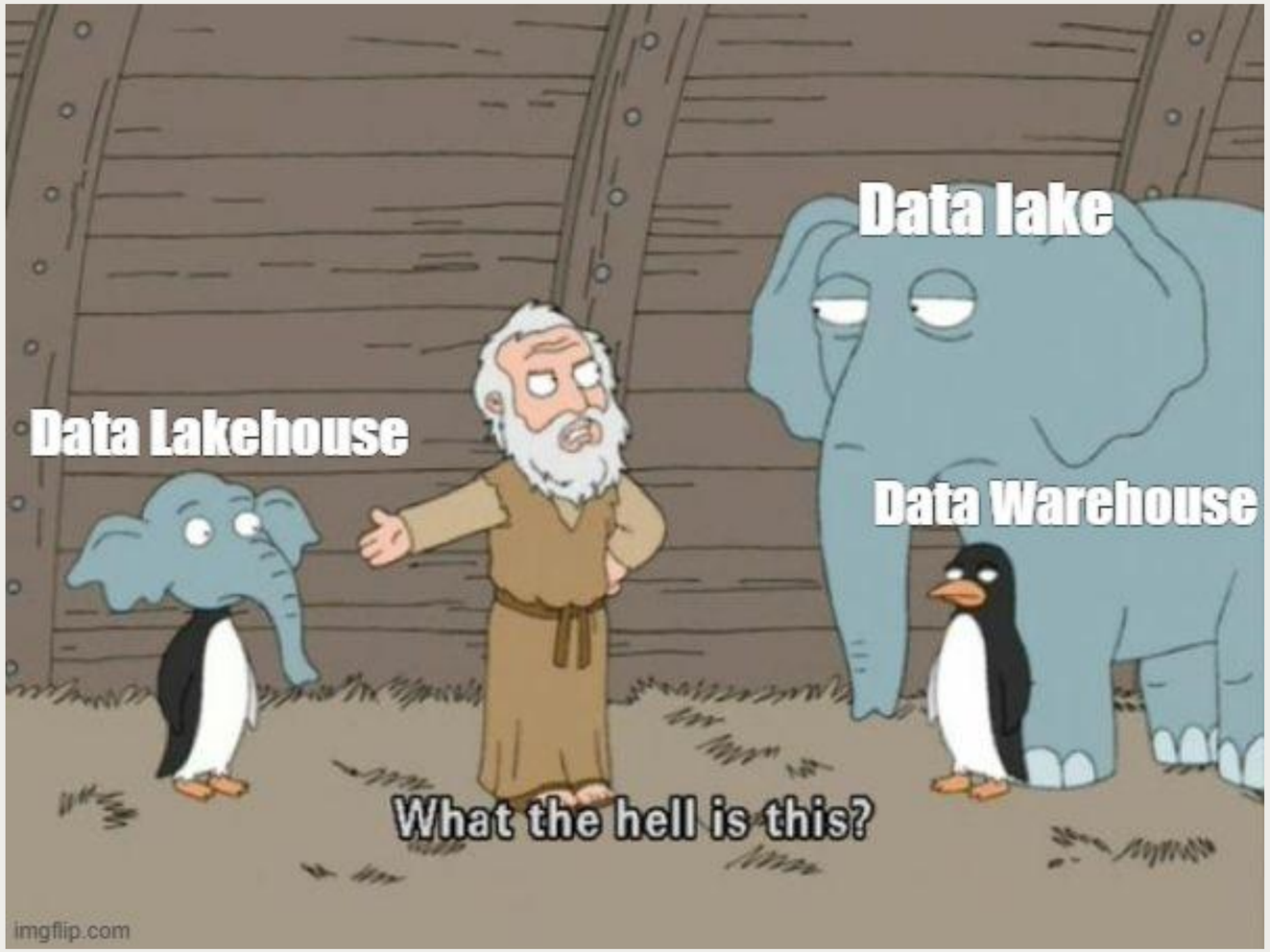


Data Storage

Feature	Data Warehous

Data Lakehouse





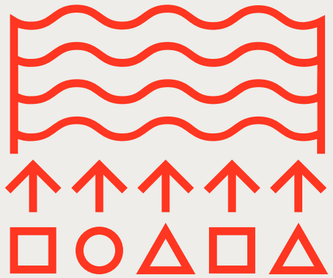
Our Data Lakehouse



+



+





Unity Catalog

- Governance
- Security

Delta Lake

- Reliability
- Structure

Data Lake (ADLS GEN2)

- Scalability
- Storage

Delta Lake key features

adds a transaction log (a _delta_log folder) next to data files.

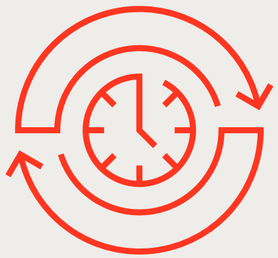
ACID Transactions

Atomicity Consistency
Isolation Durability



Data is never Lost

Version History for Data



Schema Enforcement

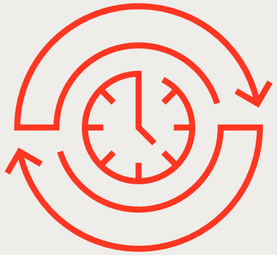


Unity Catalog key features

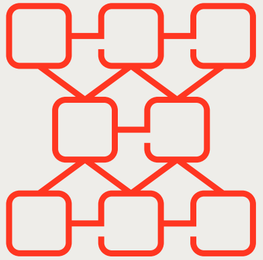
Access Control



Data Lineage



Tagging & Auditing

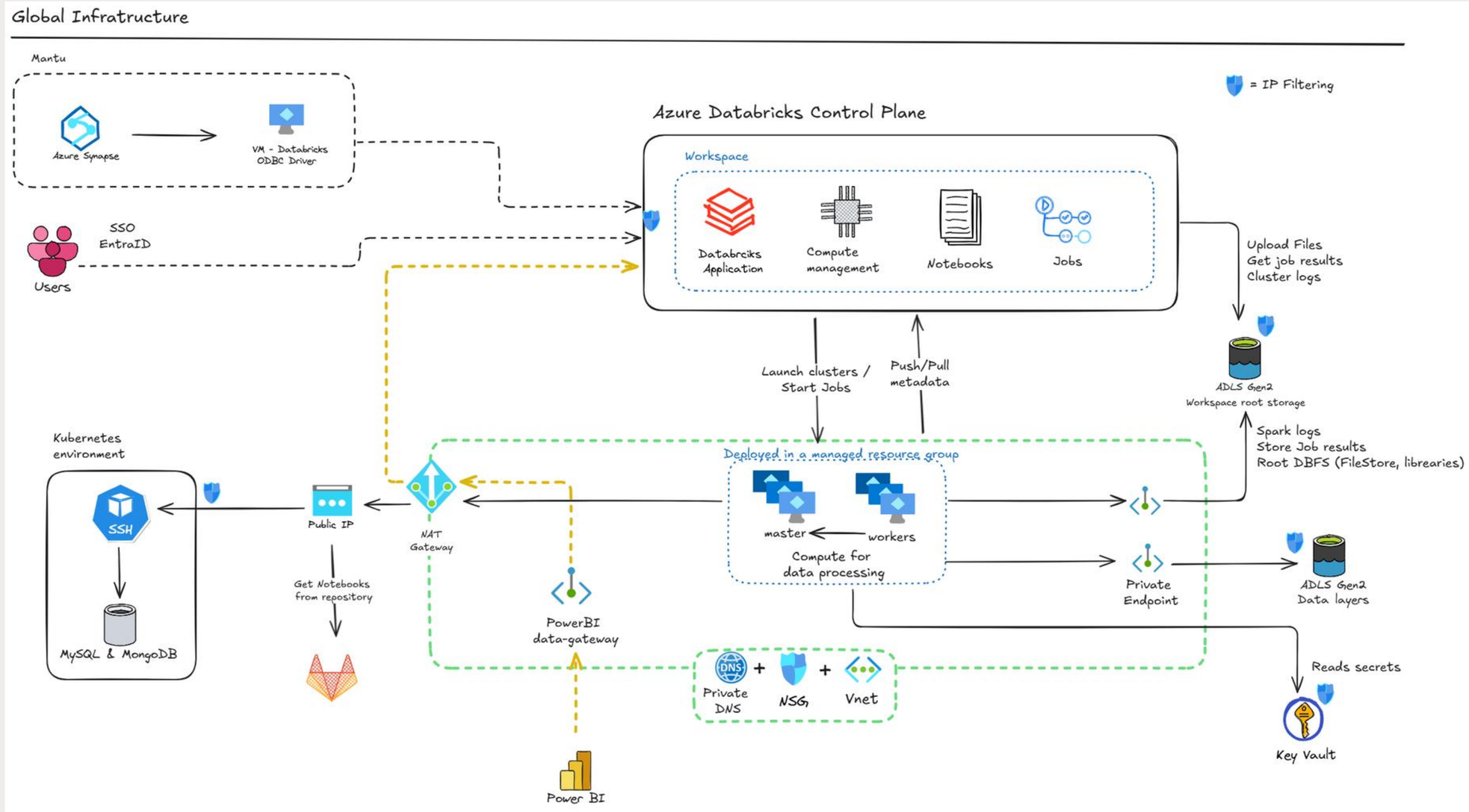


Data Storage

Feature	Data Wareho
Data Type	Structured O

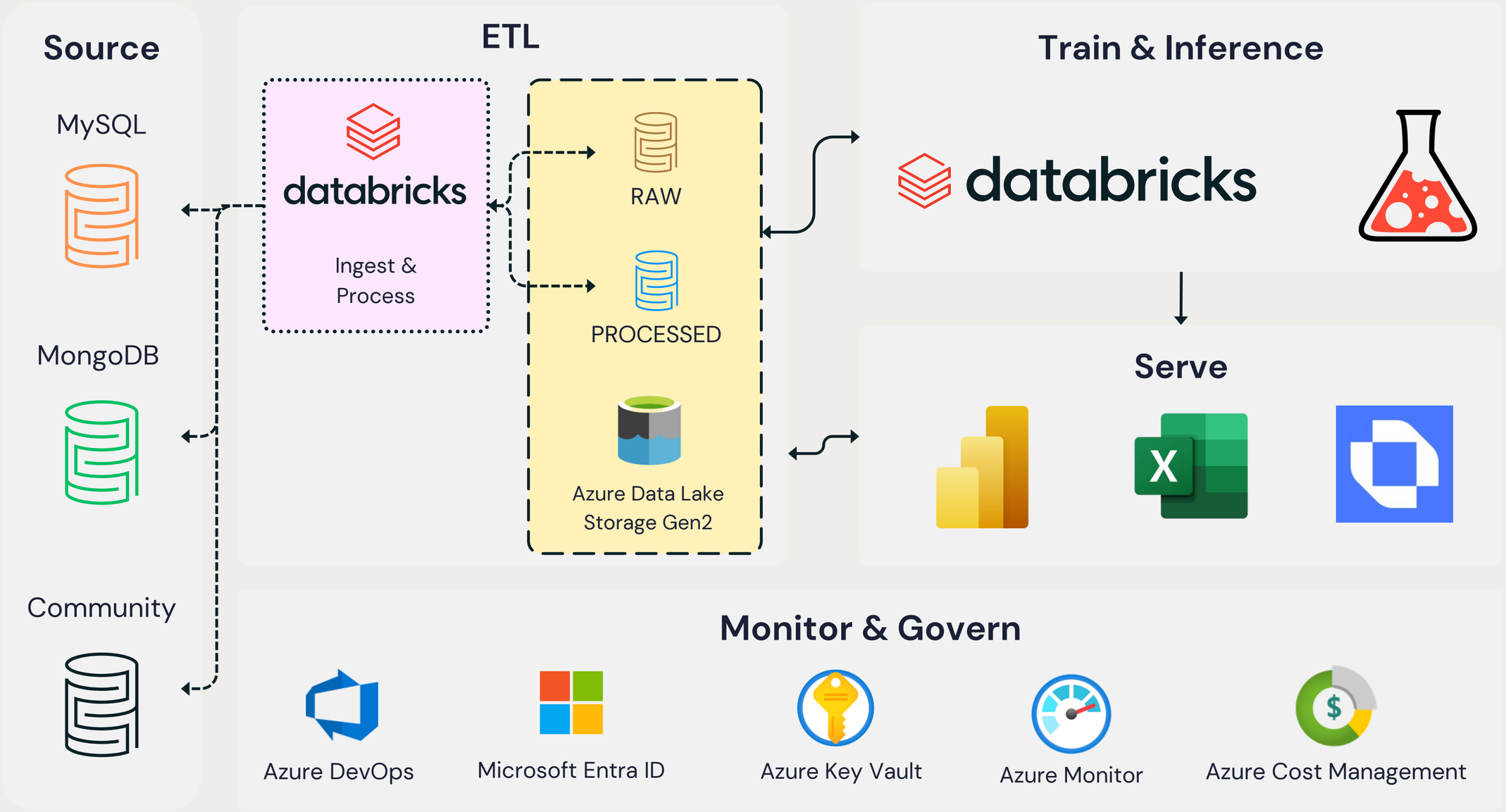
2. Architecture and key Components

Full Architecture



2. Architecture and key Components

Simplified Architecture



Data Sources





Type	Rela

Processing Engine



open-source engine designed for processing massive amounts of data incredibly quickly

Super Fast



100x faster than older systems

Massively Scalable

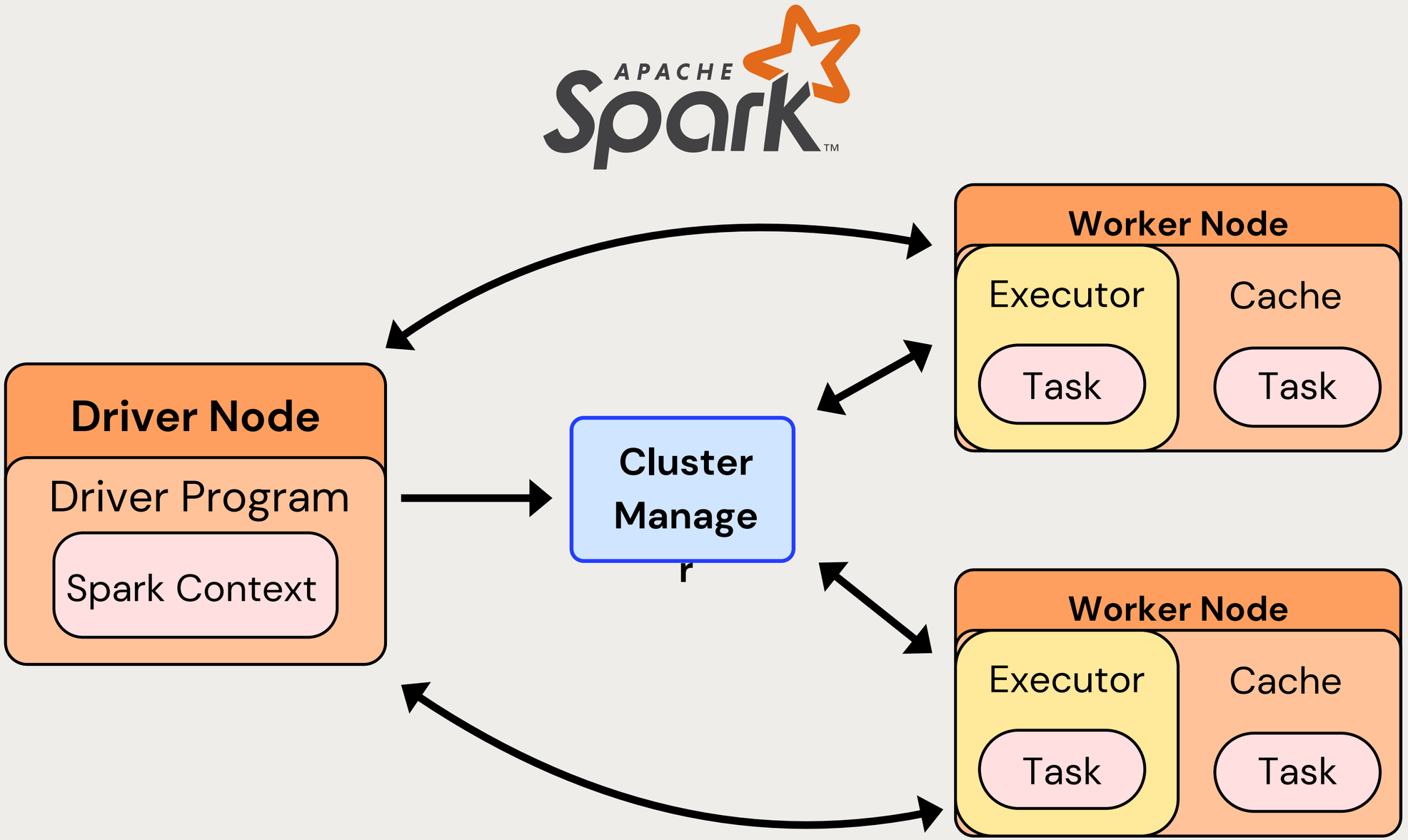


Distributes a single big job into many small tasks that run on a cluster

Speaks Many Languages



Python (PySpark), SQL, Scala, and R


APACHE
Spark™
Worker Node
Executor
Task
Cache
Task
Driver Node
Driver Program
Spark Context
Cluster Manage r
Worker Node
Executor
Task
Cache
Task

How do we load the data from on-premise sources into our new cloud Lakehouse and perform operations on it?





3. Data Operations

1.

Configure the platform

2.

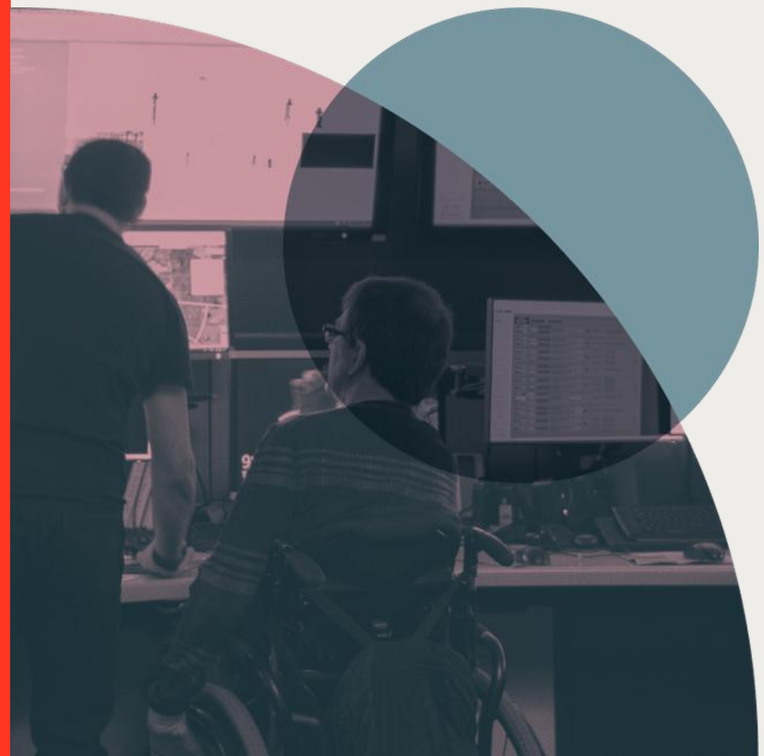
Connect to the Sources

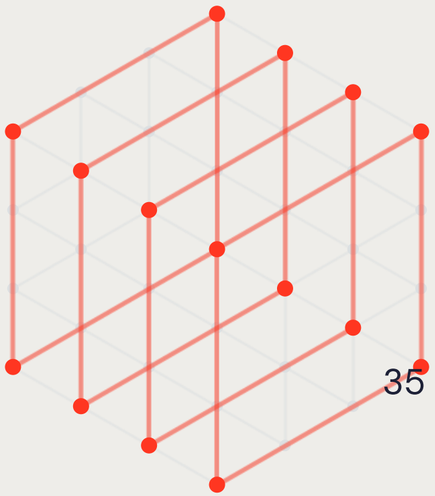
3.

Extract Data

4.

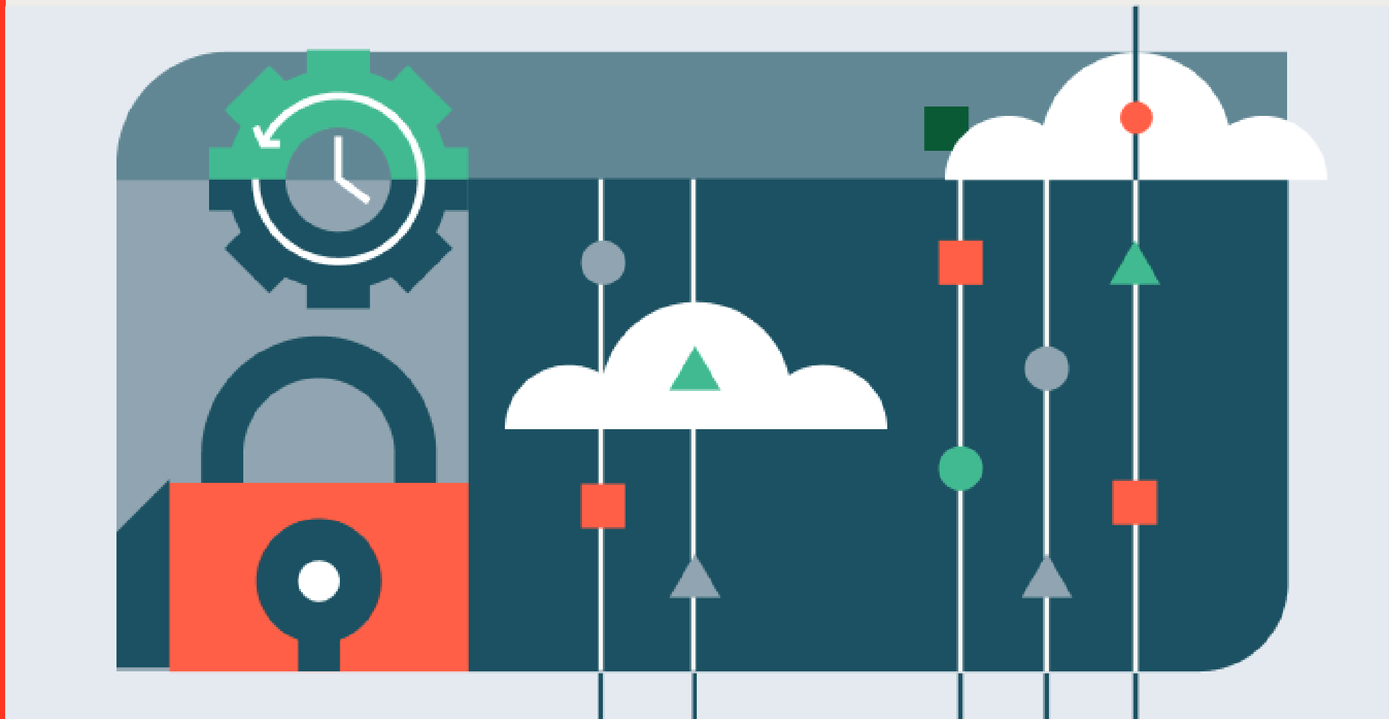
Perform Data Operation

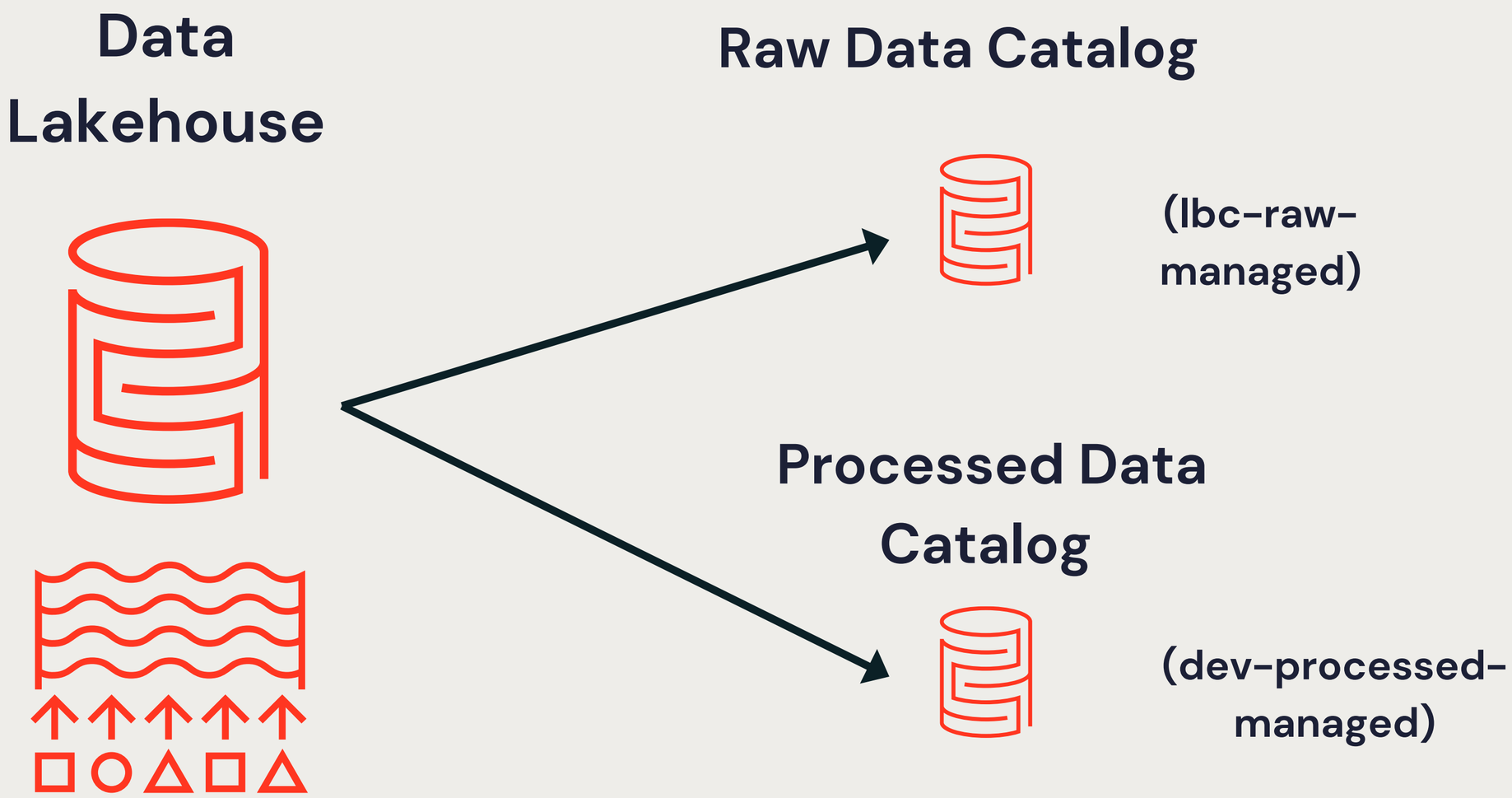




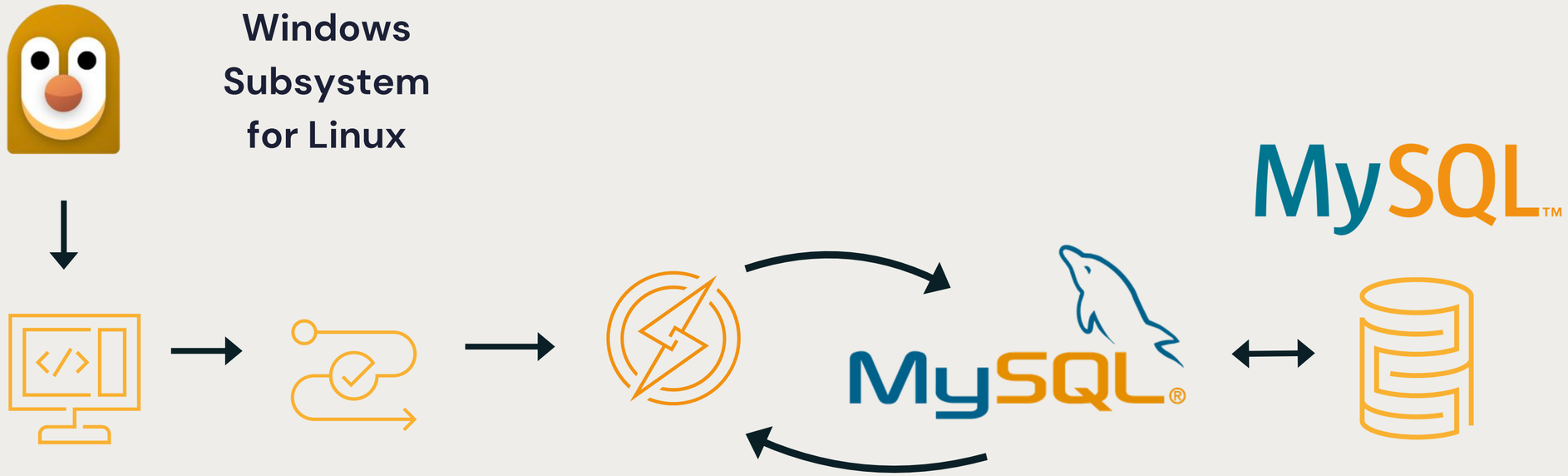
SCRUM Methodology

10 Sprints

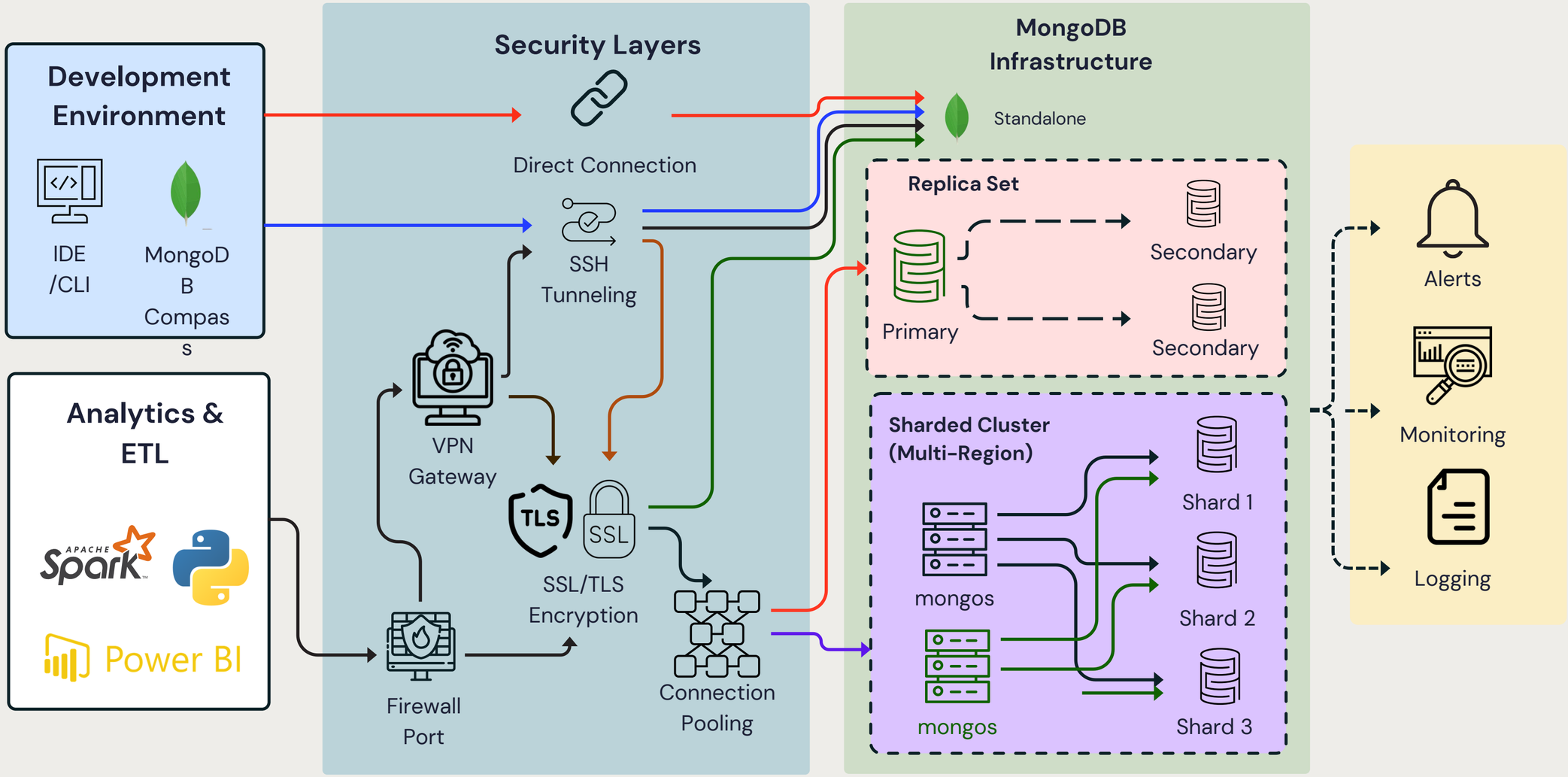



Data Lakehouse
Raw Data Catalog
(lbc-raw-managed)
Processed Data Catalog
(dev-processed-managed)

Simplified MySQL Connection

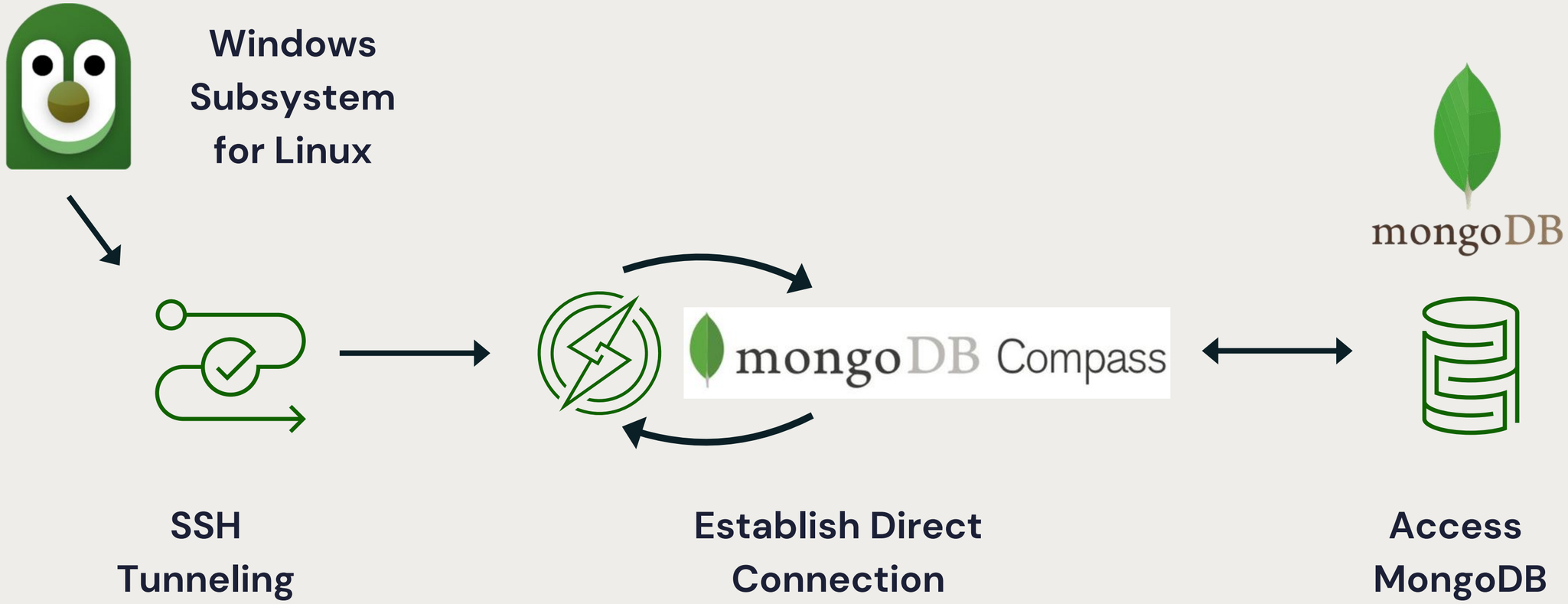


MongoDB Connection



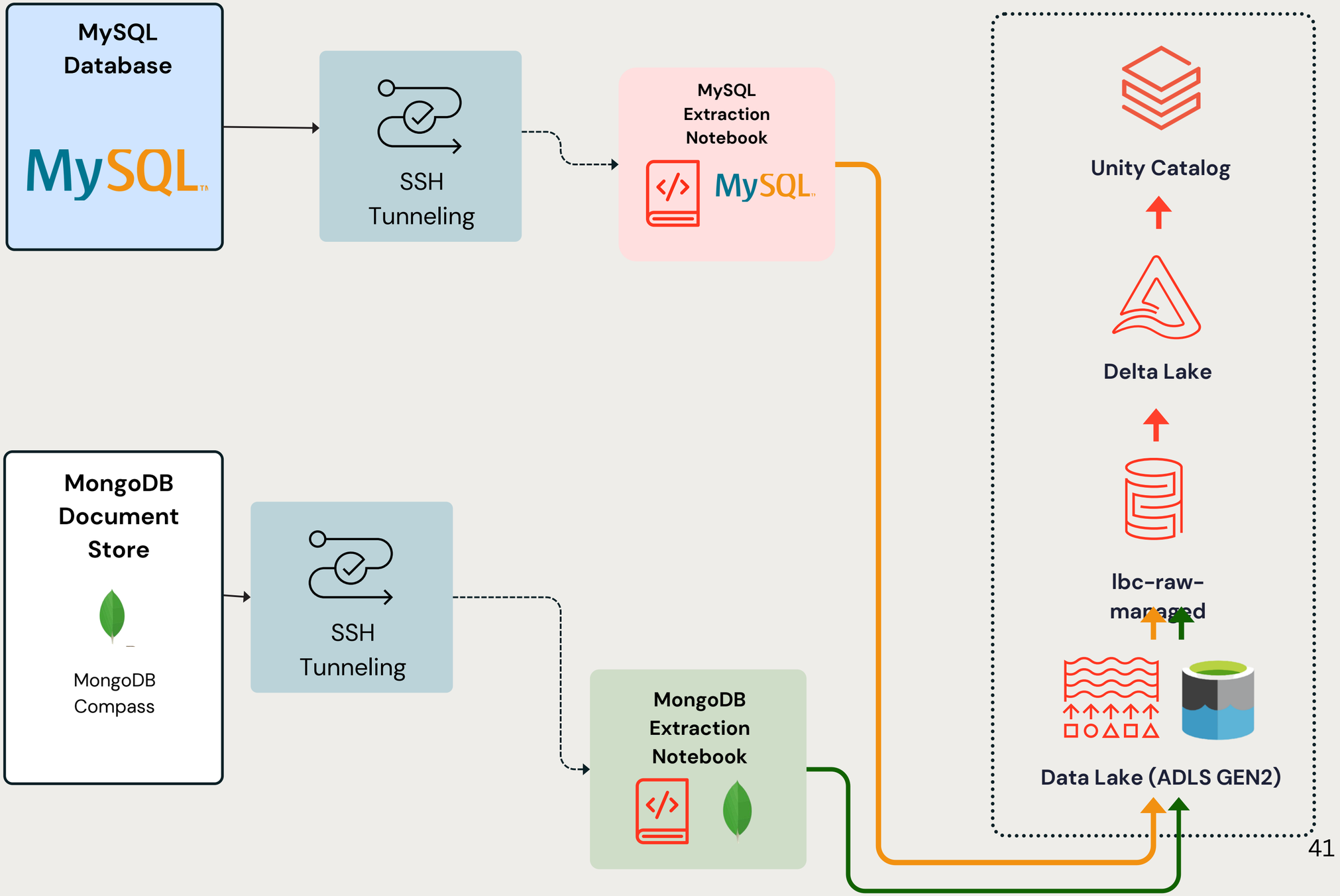
3. Data Operations

Simplified MongoDB Connection



3. Data Operations

Simplified Data Extraction



Why unit tests ?

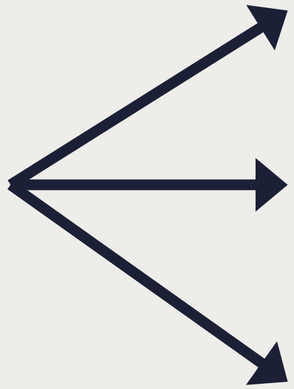
- Not enough for the code to work once



- Catch Bugs Early

Cod





Reliable

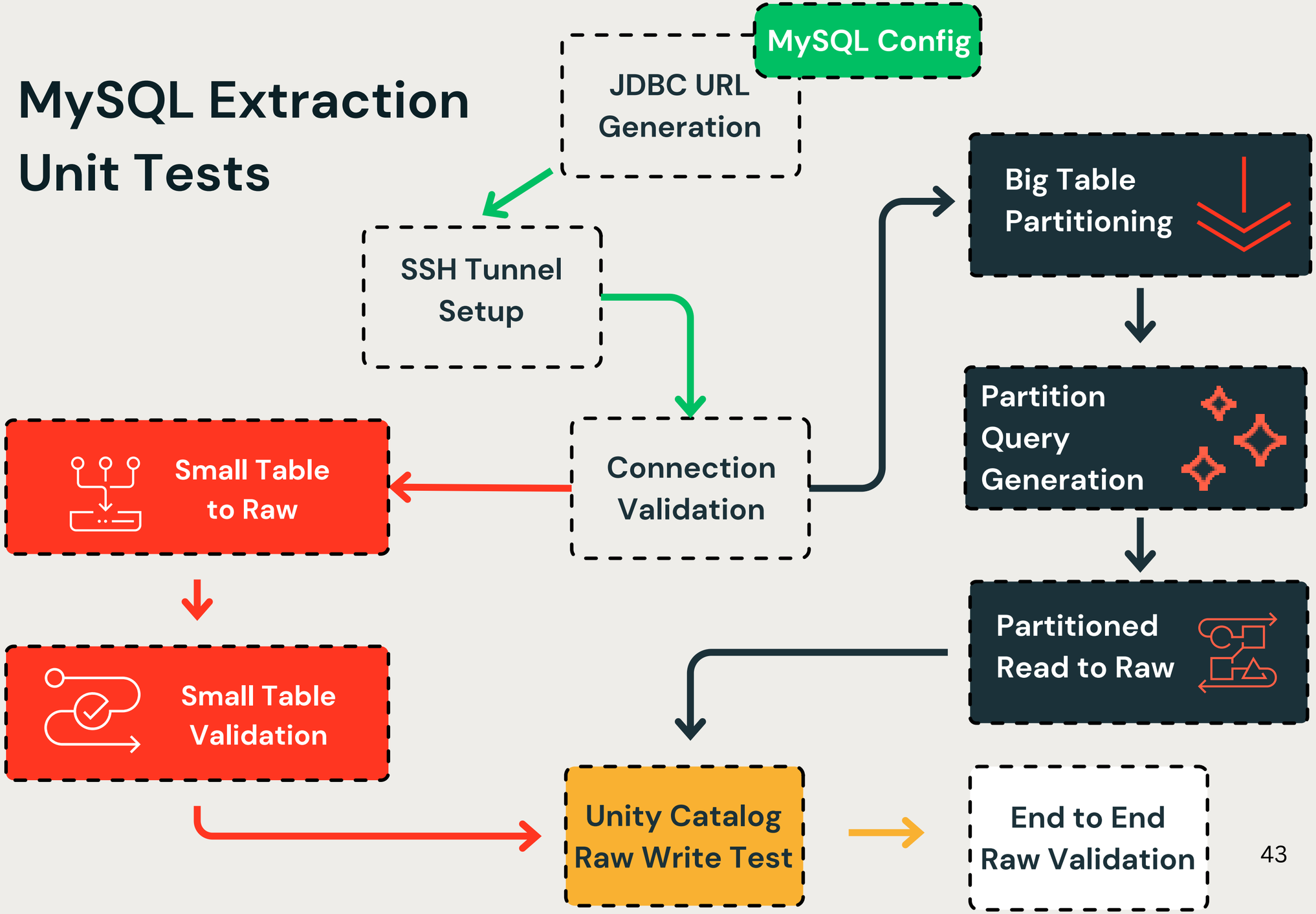


Easy to maintain Enterprise-grade

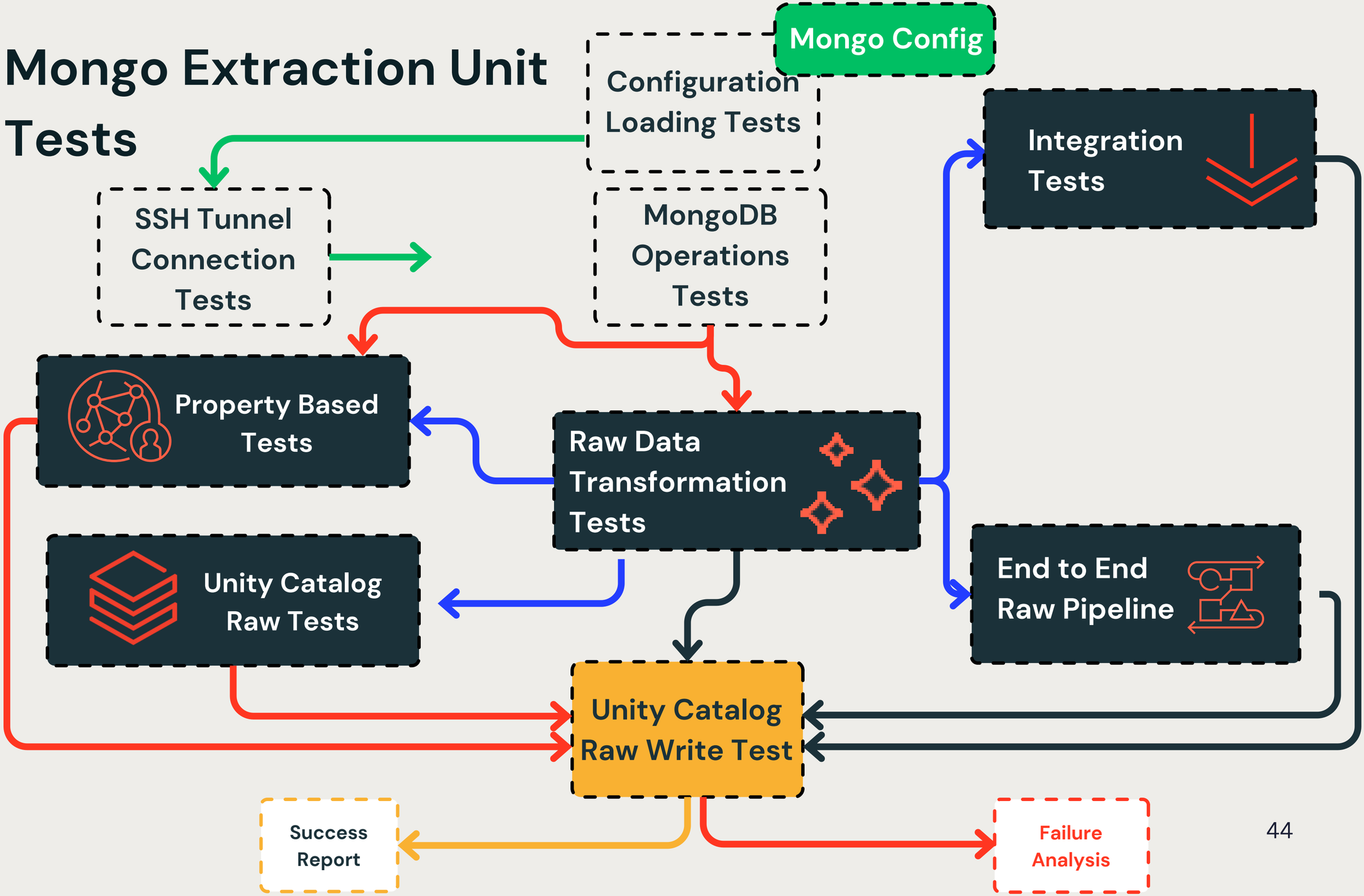




MySQL Extraction Unit Tests

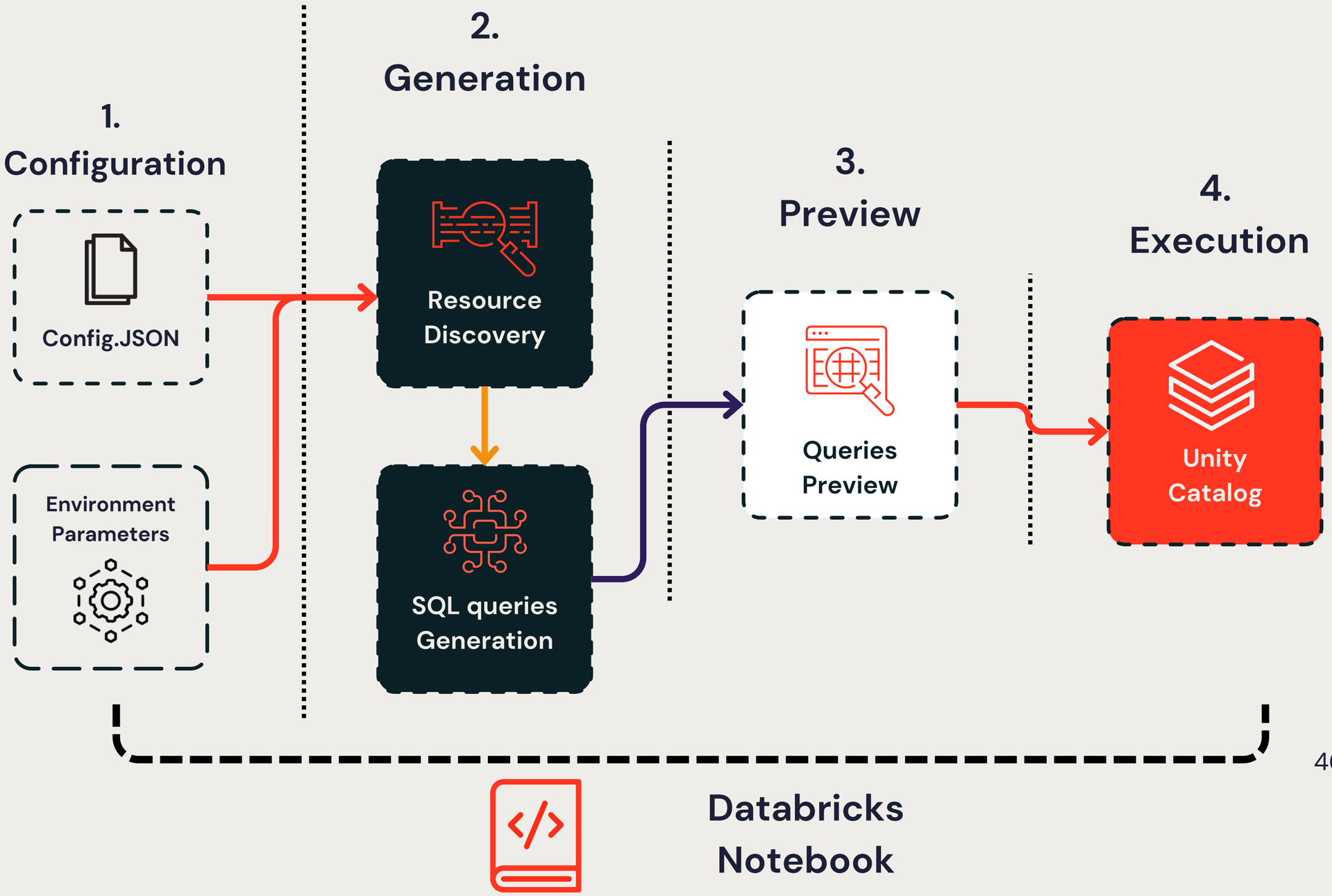


Mongo Extraction Unit Tests



Unit Tests

Component	Testing Foc
Configuration Management	JSON config secret valid
SSH Tunnel Operations	Secure conn establishme
MongoDB Connectivity	URI generati collection re

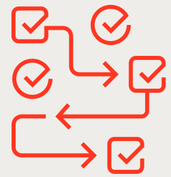

1.
Configuration
Config.JSON
Environment
Parameters
2.
Generation
Resource
Discovery
SQL queries
Generation
3.
Preview
Queries
Preview
4.
Execution
Unity
Catalog
Databricks
Notebook

Access Management

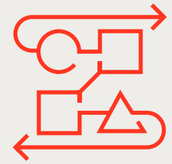
1. Basic	2. Intermediate	3. Advanced	4. Full
Catalog Usage CREAT E	**Schema** Usage SELEC T	**Tables** SELEC T INSERT	**All** Bulk Acces s

Need to perform Automated Modification :

GDPR Right of Deletion of a company



Company wants to bulk modify its Cards or Information



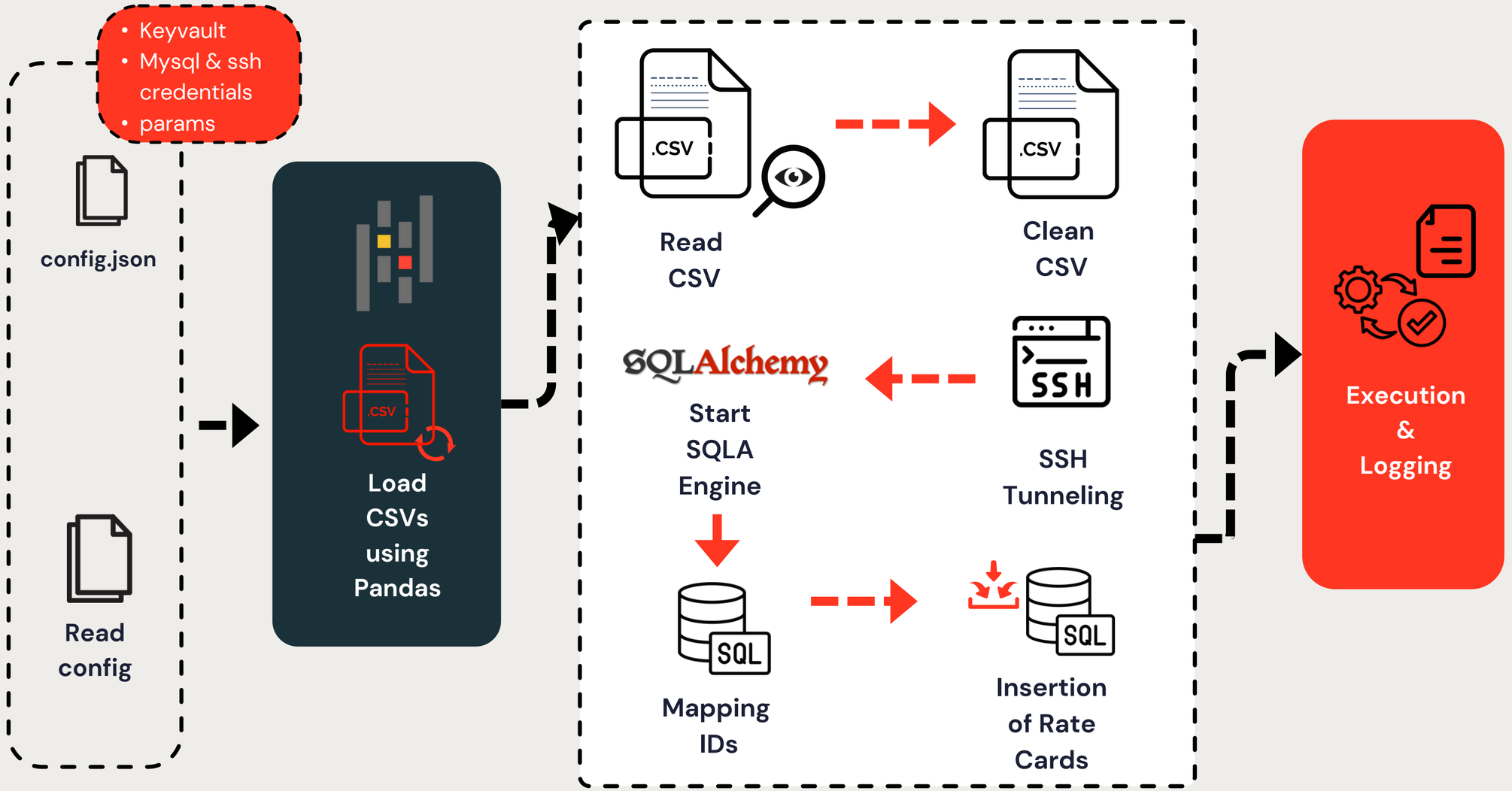
Company wants to submit a large number of RFP



3. Data Operations

Automated Data Modification

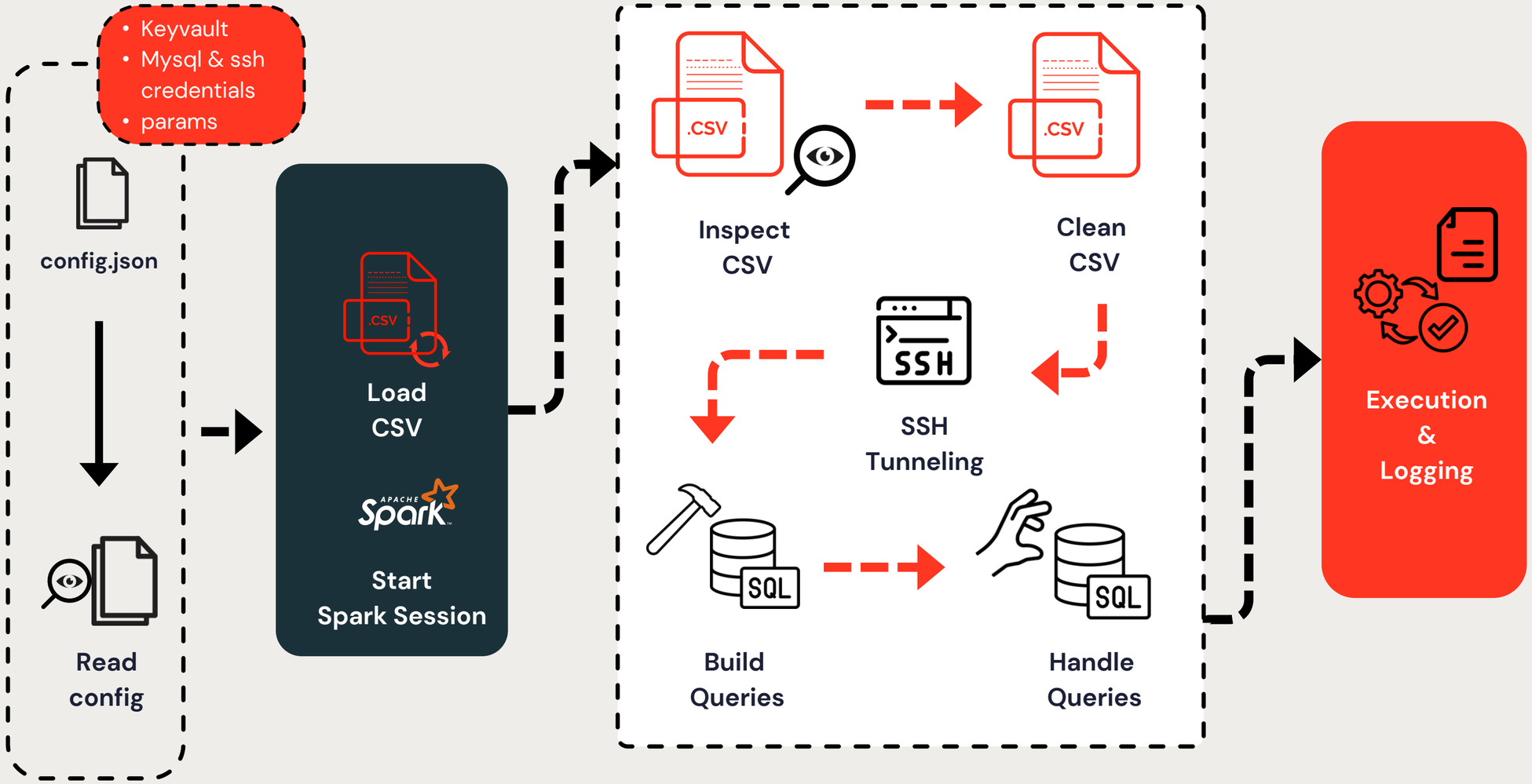
Rate Cards Insertion



3. Data Operations

Automated Data Modification

Rate Cards Deletion



Company wants to create a large number of RFPs



Manual

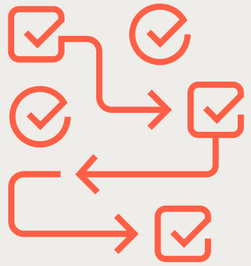
- Time Consuming
- Error Prone



Robotic Process Automation (RPA)

- Time Consuming
- Not Scalable
- Requires a lot of computatinal power

Company wants to create a large number of RFPs

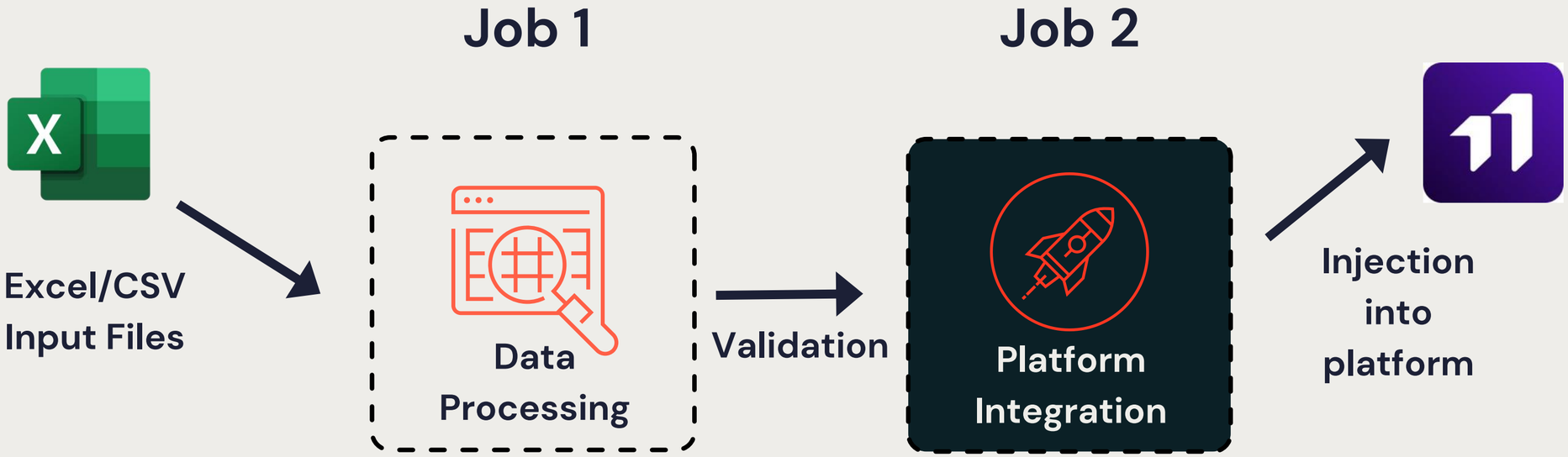


API Based Operations

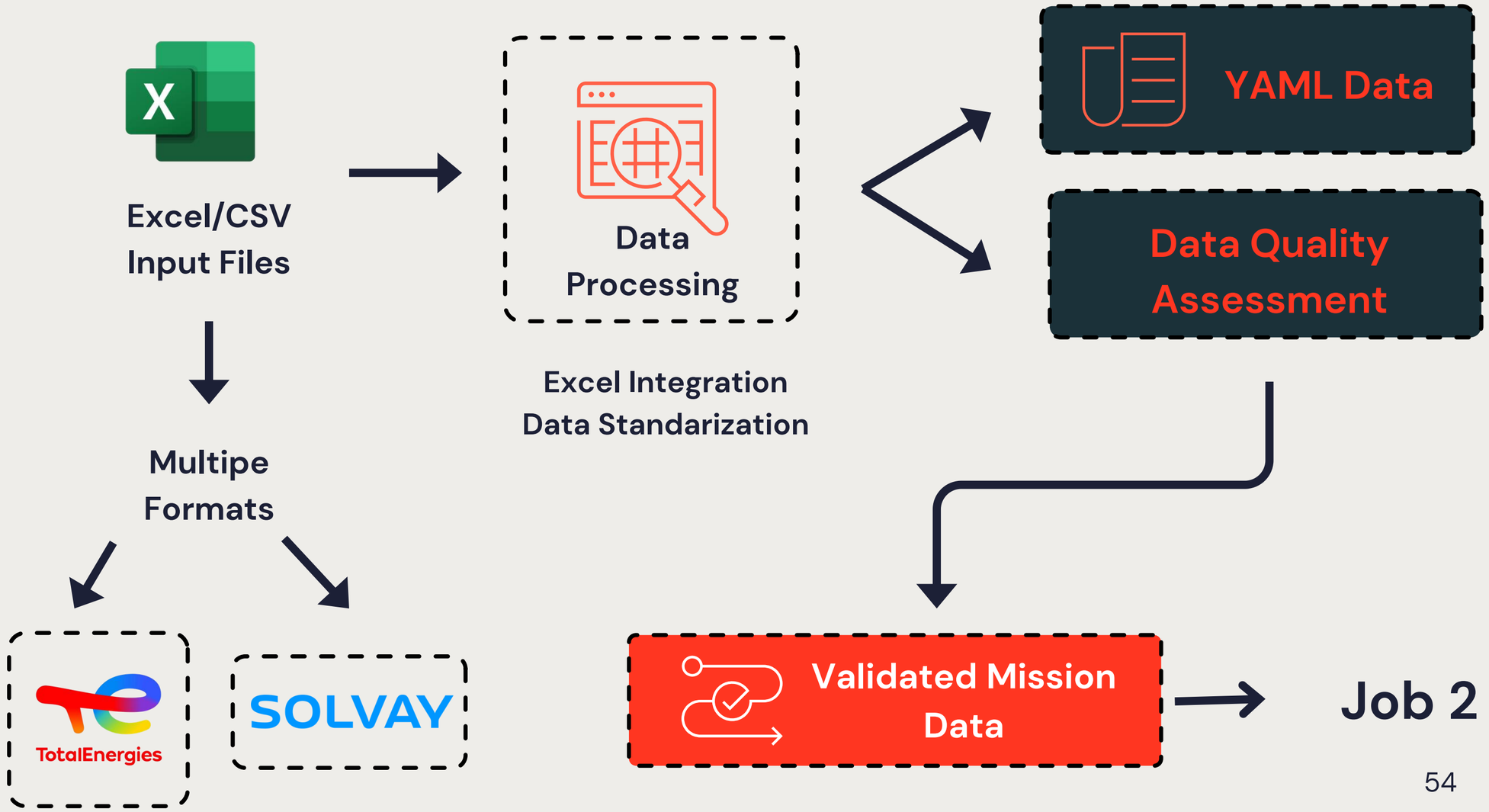
- Scalable
- Secure
- Efficient
- Fast

POSTMAN

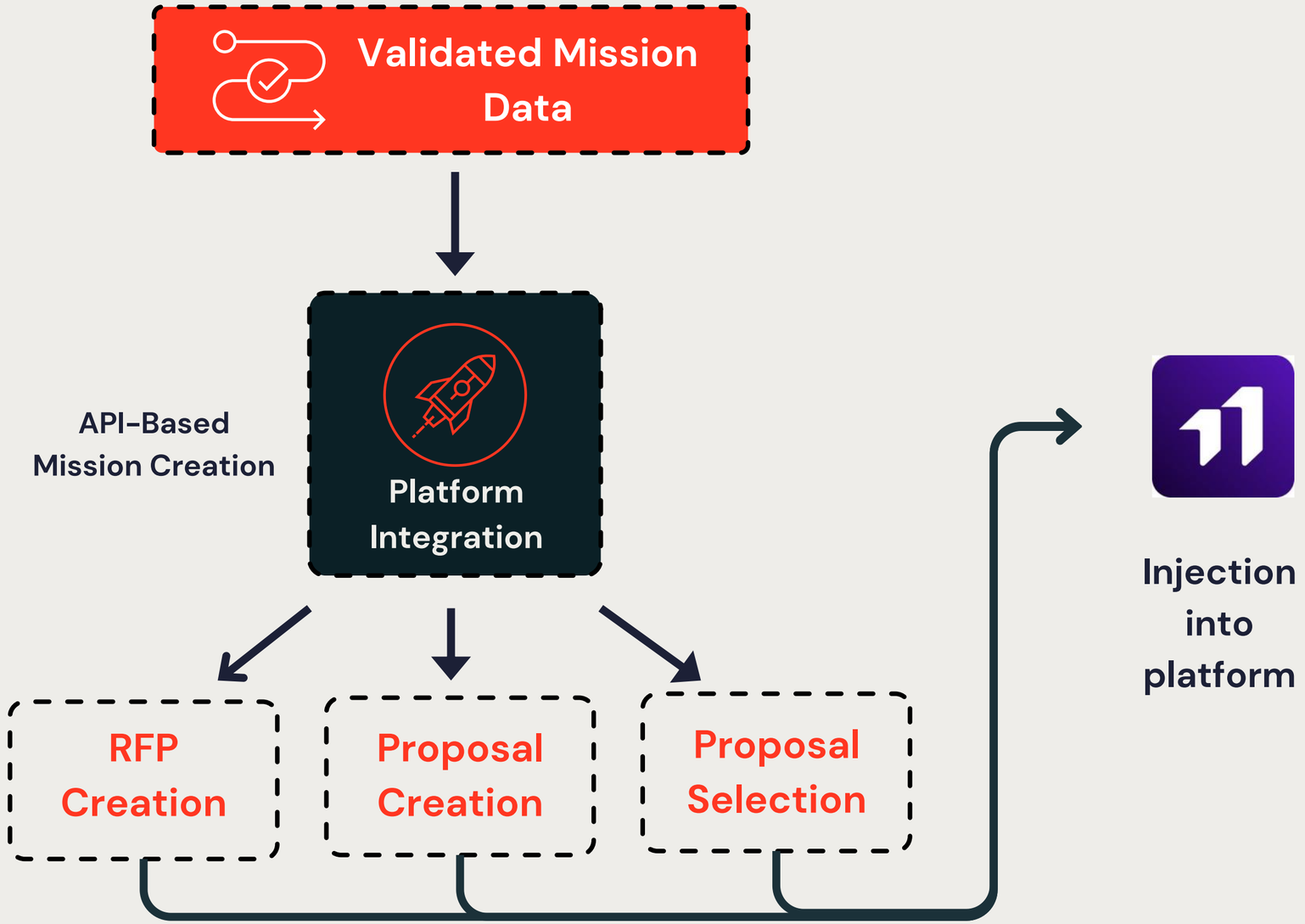
Simplified Workflow

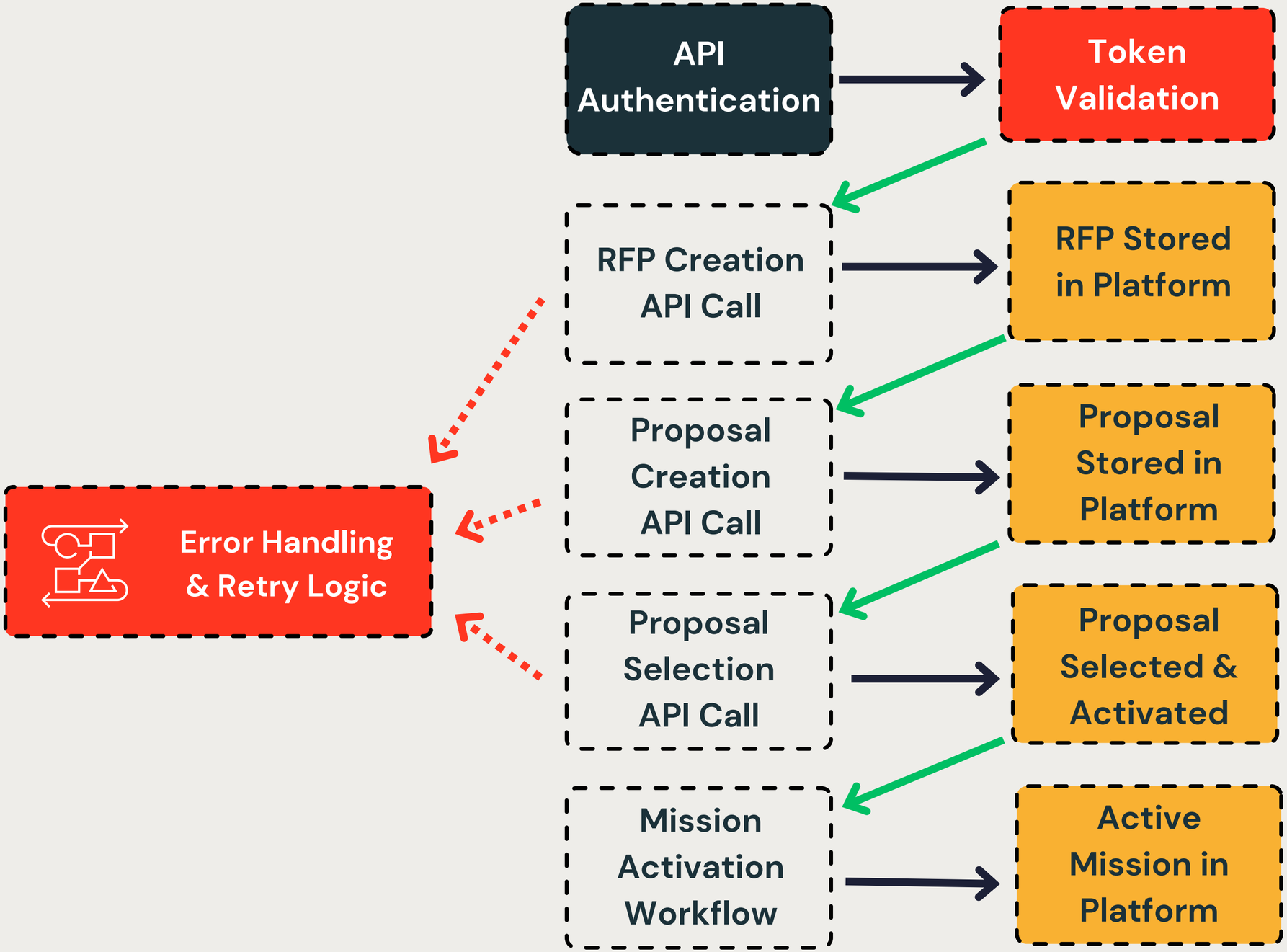


Job 1

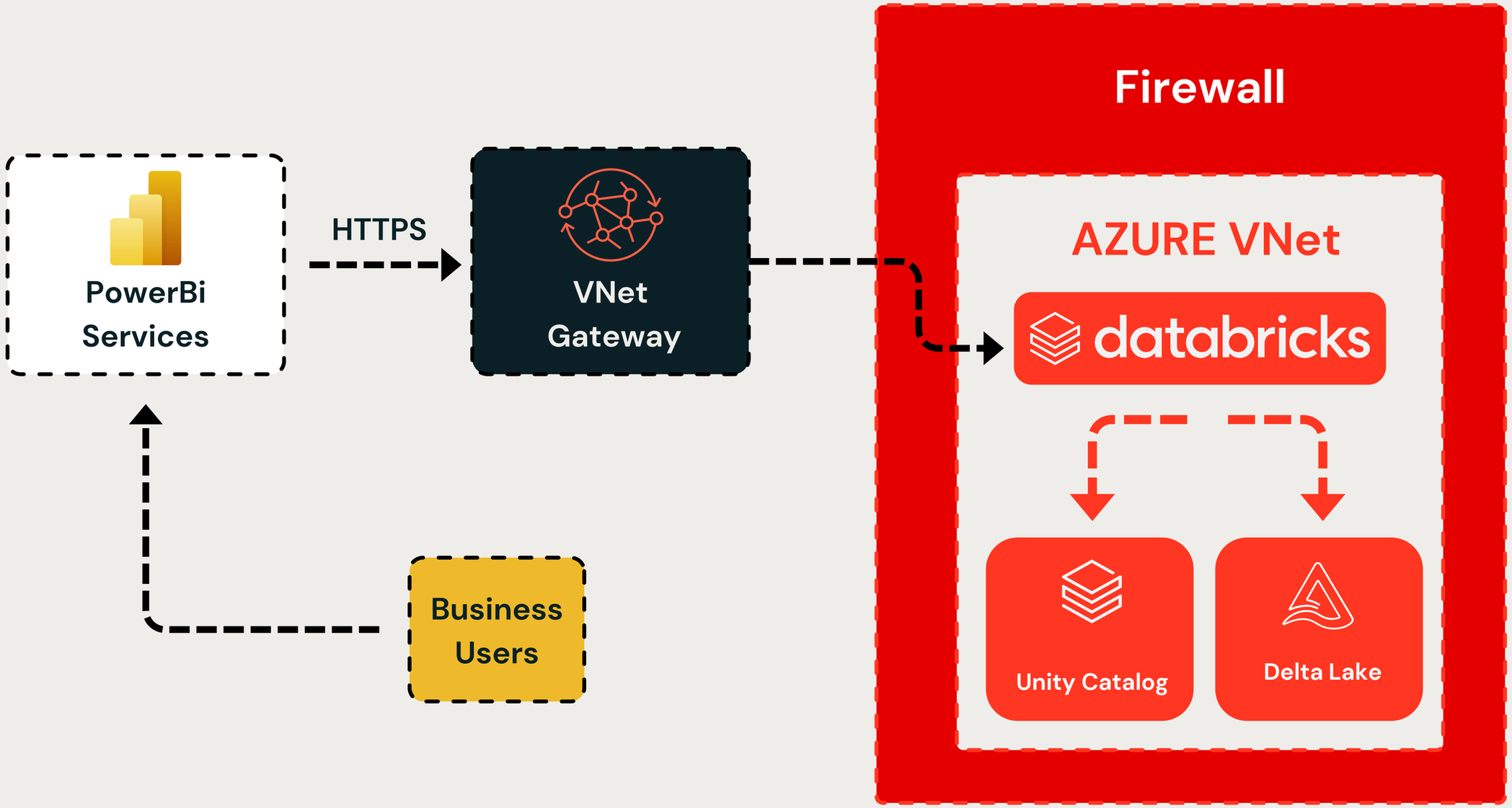

Excel/CSV
Input Files
Data
Processing
YAML Data
Data Quality
Assessment
Excel Integration
Data Standarization
Multipe
Formats
TotalEnergies
SOLVAY
Validated Mission
Data
Job 2

Job 2


Validated Mission Data
API-Based Mission Creation
Platform Integration
RFP Creation
Proposal Creation
Proposal Selection
Injection into platform


API Authentication
Token Validation
RFP Creation API Call
RFP Stored in Platform
Proposal Creation API Call
Proposal Stored in Platform
Error Handling & Retry Logic
Proposal Selection API Call
Proposal Selected & Activated
Mission Activation Workflow
Active Mission in Platform

Power BI Integration



Power BI Integration



Financial Analytics Dashboard

Operational performance Dashboard

Legal and compliance Dashboard

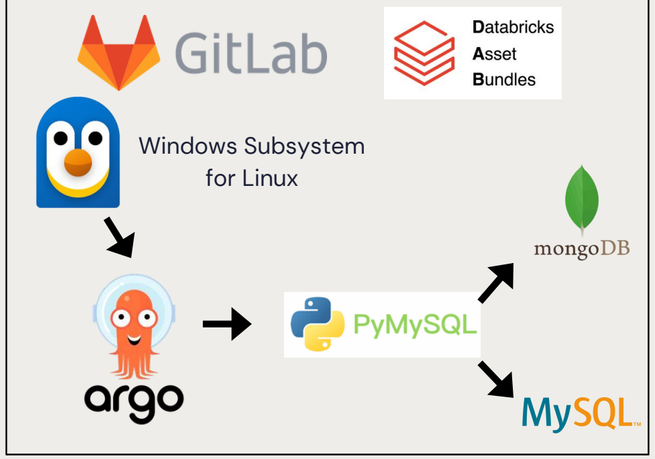
3. Data Operations

Tools Used

Tooling & Collaboration



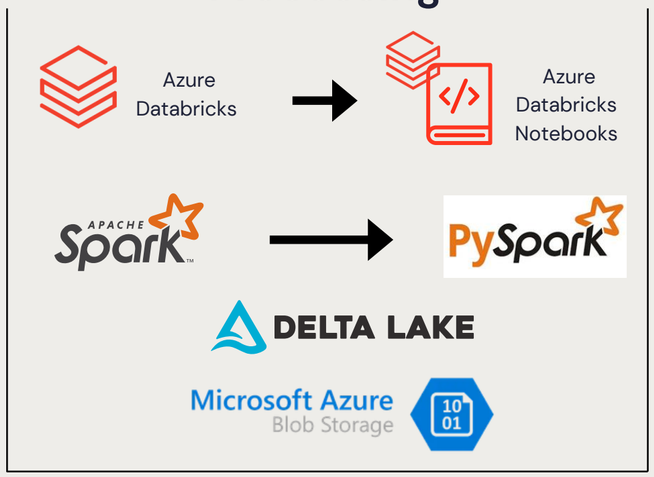
Development & CI/CD



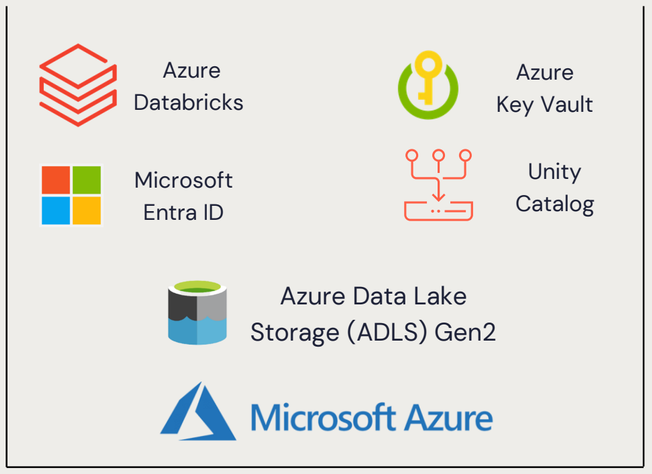
Data Transformation & Analytics

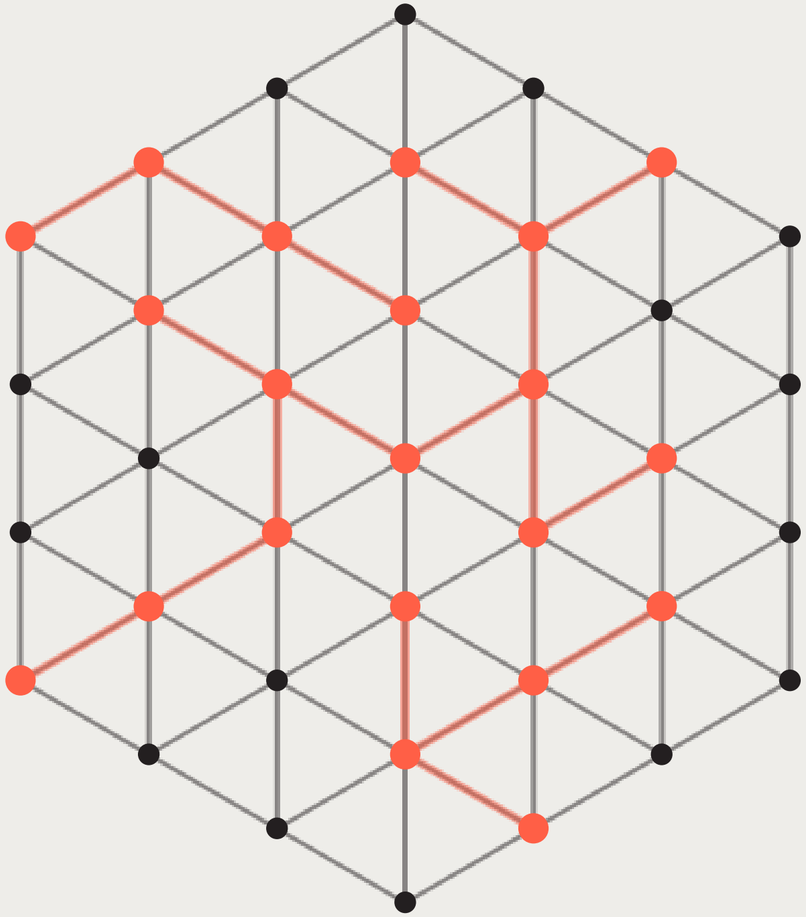


Data Engineering & Processing

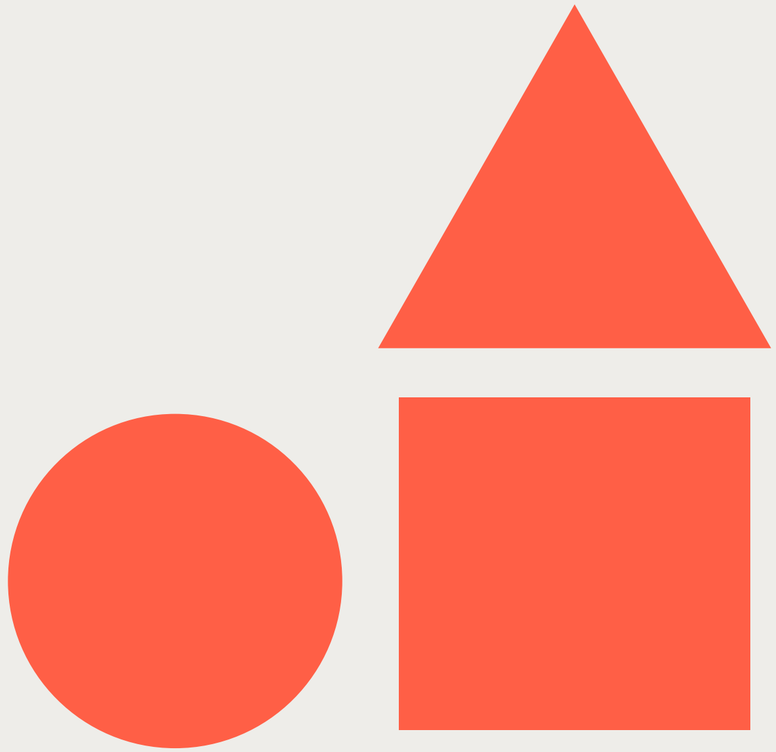


Cloud & Platform Services





DEMO



4. Value and Impact

GDPR Tag Coverage



70%

Manual

→

100%

Automated

Schema Provisioning Time



1–2 Hours
Manual

→

2–5 Minutes
Automated

MySQL



3 million Records

50 Minutes

MongoDB



>100k Records

2 Minutes



Paralle

MySQL



MongoDB



3x Day

Databricks + asset bundles



100% Success Rate

↓

Fresh data every 8 hours

Fresh data every 8 hours

Serve →

PowerBi Services

Constant

Dashboards update



↓



More Accurate

Dashboards

Reduce *Manual* Generation Time

16hrs



↓

(One Financial Report)

4hrs

BI Results and Evaluation

Fresh data every 8 hours

Serve →

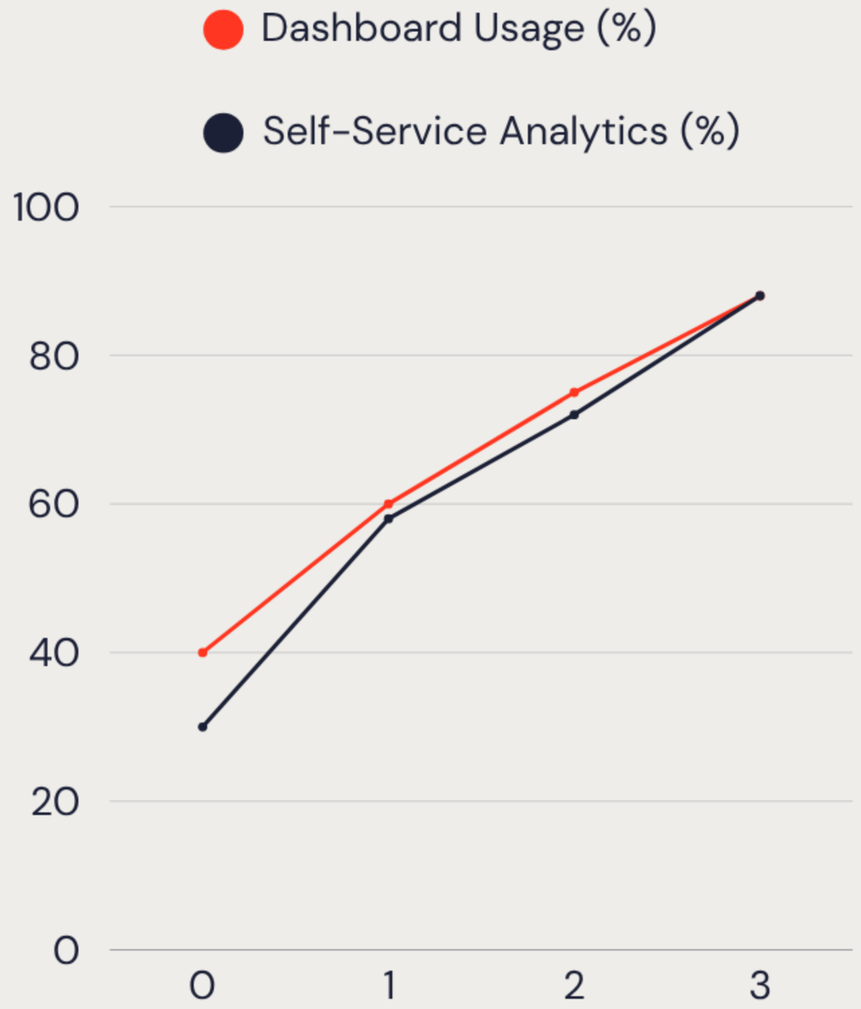
PowerBi Services

Number of **new** Custom Reports (/Month)

5

↓

20



Fresh data every 8 hours → Serve → PowerBi Services

Financial Analytics Dashboard

- Revenue tracking
- Payment monitoring

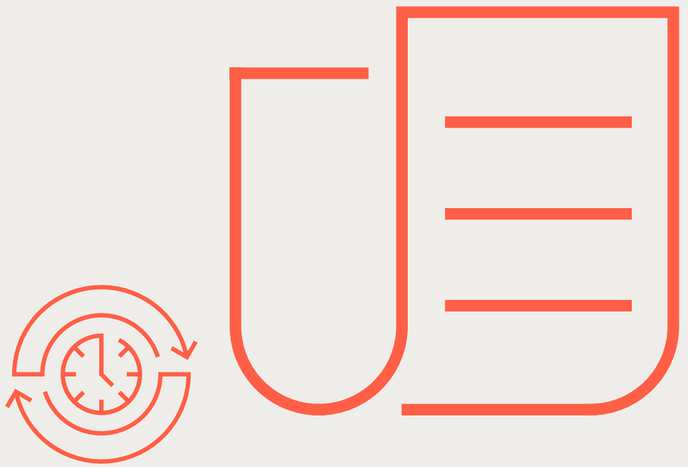
Operational performance Dashboard

- Mission completion rates
- Consultant performance indicators

Legal and compliance Dashboard

- Contract tracking

Processing Time (100 records)



6->8 Hours
Manual

→

<5 Minutes
Automated

With CSV provided

Audit Capability

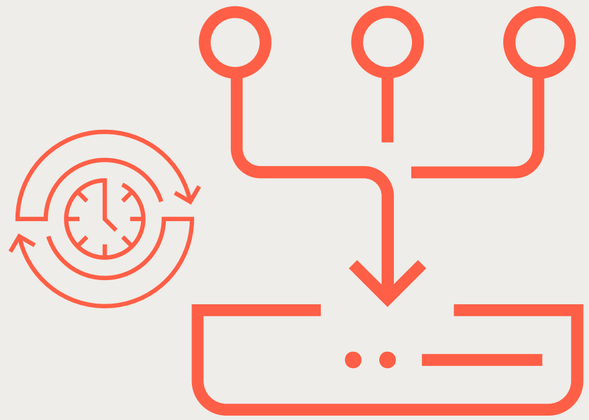


Screenshots
Manual

→

Complete Logging
Automated

Processing Time for injections (30 Mission)



2 business Days

Manual

→

90 Minutes

Automated

4. Value and Impact

Platform Transformation

Fragmented Data

Inconsistent Data

Inefficient Workflow

Lack of Scalability

Poor Data Governance

Unified cloud-native Data Lakehouse

Efficient Scalable Pipelines

Automated ETL Process

Secure governed database

Delta Lake

ADLS Gen2

APACHE Spark

databricks

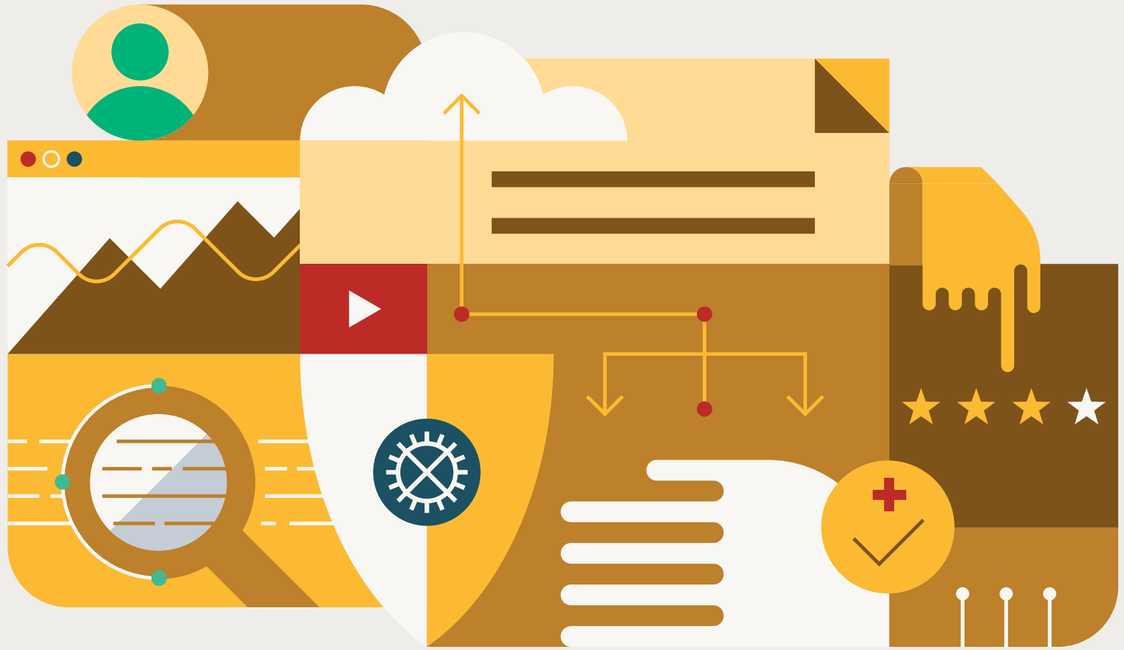
Unity Catalog

Microsoft Entra ID

Platform Impact

Automated ETL Process

Operational Excellence



Frees teams to focus on strategic initiatives instead of maintenance.

Centralized governance with automated data lineage and access controls.

Regulatory Compliance

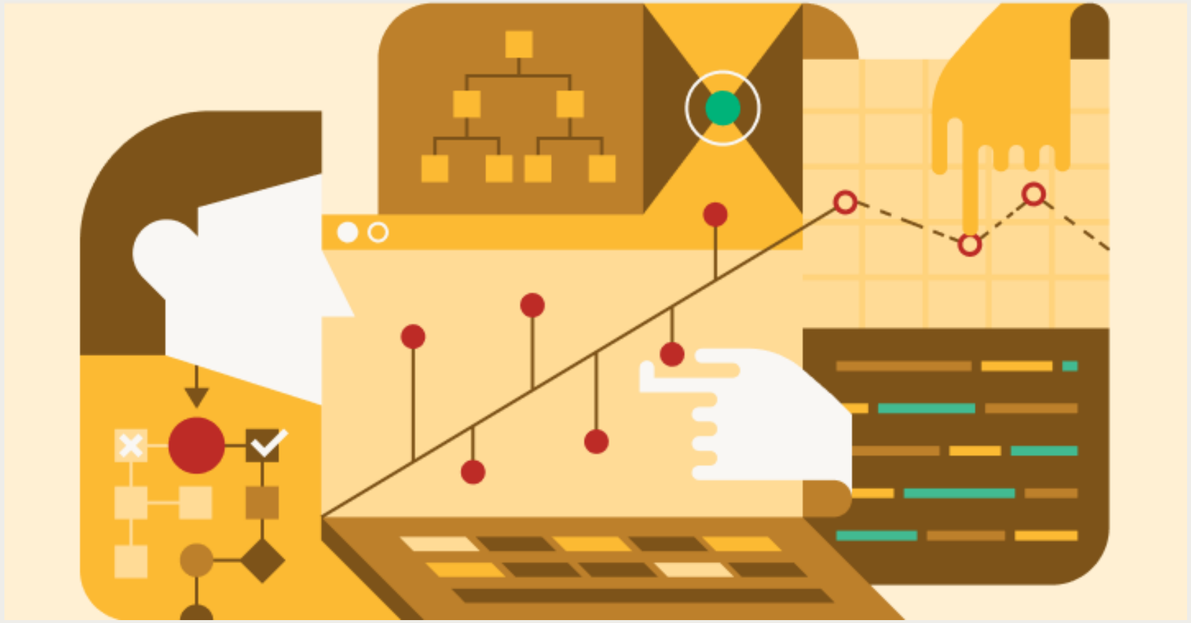


Reduces compliance risk (e.g., GDPR)
and simplifies audit responses

Platform Impact

Robust platform ready for advanced analytics and AI

Foundation for Innovation



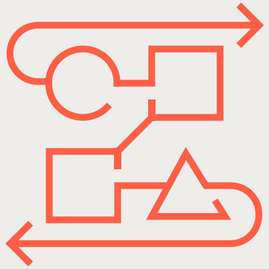
Unlocks potential for future high impact projects like machine learning models.

Platform Limitations

Real time Processing



Self Service Capabilities







5. Conclusion & Perspectives

Transformed a fragmented, manual system into a unified, automated data platform.



DELIVERED Significant, measurable gains in efficiency, governance, and scalability.



A robust and trustworthy foundation for all future AI at Little Big Connection.

5. Conclusion & Perspectives

Clean

+

Reliable

Dat

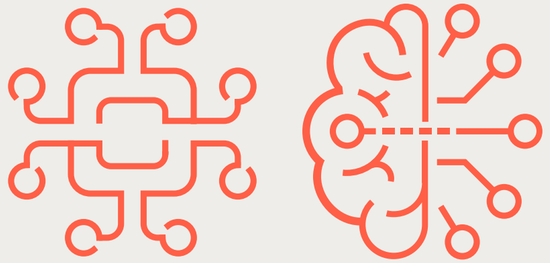
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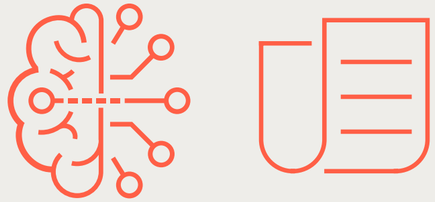
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Enterprise grade AI models





5. Conclusion & Perspectives

Critical by Q4 2025	Real time Processing	
High by Q1 2026	AI generated Reports	
Medium by Q3 2026	API Integration Expansion	

"The goal is to turn data into information, and information into insight."

– Carly Fiorina



LittleBig Connection

This is the robust foundation that will power the future of AI and Data Science at LB.

Thank you all for your time and attention.

